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EvoRiri: Interpretable Evolutionary Search for Diagnosing Reward-Design Failures in a Bangla Mental-Health Assistant

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Abstract

Mental-health conversational systems require evaluation that considers empathy, risk-aware safety, and actionable response structure, yet optimising hand-designed proxies for these qualities can introduce reward-design failures. We present EvoRiri, an interpretable evolutionary-search framework for a template-based Bangla–Banglish–English mental-health assistant. Rather than updating language-model parameters, the method searches a compact mixed continuous–discrete genome controlling response composition, escalation behaviour, structure, and template configuration. A scalar fitness function combines automatic empathy, safety, and structure proxies with per-example safety penalties. Experiments on 2,872 de-identified records represented as synthetically constructed prompts show that the reported evolved configuration achieves higher structural completeness and fitness than the fixed reference genome under a frozen scoring pipeline, while maintaining a similar escalation-calibration safety proxy. Trace-level analysis was central to the study: it identified question-only structural collapse, empathy-marker stuffing, a structure-marker coverage gap, and a response-part-budget truncation defect that aggregate fitness scores alone did not reveal. After correcting the scoring and generation pipeline, the reported configuration reaches complete measured structural coverage under the current proxy. A 12-point structure-weight sweep and five equal-budget random searches reach the same high-fitness region as the GA. Re-evaluation of archived GA and random-search genomes reveals no resolvable difference between the methods under the available evaluation design. Thus, for this genome space and fitness function, the contribution is not a demonstrated unique optimisation advantage of GA search; it is an interpretable framework for exposing how reward design, thresholded generator behaviour, and baseline construction shape observed performance. The findings are limited to a synthetic single-turn corpus and rule-based proxy metrics, and do not establish clinical effectiveness or deployment readiness.

Keywords: evolutionary algorithms; genetic algorithms; mental-health chatbots; large language models; empathy; safety; multi-objective optimization; conversational agents; Bangla; risk-aware AI

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1. Introduction

Conversational agents, increasingly built on large language models (LLMs), are now widely deployed for mental-health support [1,2]. Deploying these systems safely requires

more than grammatical correctness or task accuracy; responses must be empathetic, structurally helpful, and aligned with risk-aware safety principles [3–5]. These systems are also unevenly distributed: emotion-aware AI is shaped largely by Western, educated, industrialised, rich and democratic contexts, sentiment tooling rarely extends beyond a handful of high-resource languages, and low- and middle-income countries often lack the infrastructure for AI-mediated support, a divide that risks a bifurcated system benefiting the Global North [6]. The present work targets that gap directly: a Bangla/Banglish/English assistant that runs without GPU infrastructure or API access.

Conventional fine-tuning and reinforcement learning from human feedback (RLHF) are powerful but expensive as it involves significant computational demands that may require considerable investments in infrastructure and resources [7]. In contrast, evolutionary algorithms (EAs) can explore large design spaces and evolve novel high performing solutions being both lightweight and scalable [8]. A growing literature applies them in combination with LLMs [9,10]. In the context of conversational agents, this makes them attractive for optimizing template-based generators or routing strategies without modifying, or in the present case without invoking, an underlying language model [11].

Unlike prior evolutionary prompt optimization approaches that focus primarily on reasoning accuracy, feature activation, or prompt engineering, the proposed framework focuses on optimizing emotionally sensitive conversational behavior under strong safety penalty terms. The system introduces risk-aware escalation, progressive structure enforcement, and trace-level behavioral analysis for a multilingual mental-health assistant. To the best of our knowledge, this is among the first studies applying interpretable genetic optimization to empathy–safety–structure balancing in Bangla mental-health conversational systems.

In this article we introduce a Genetic Algorithm (GA)-based risk-aware framework for *Riri*, a mental-health-oriented assistant designed for Bangla-speaking users. The system composes responses via a structured, template-based generator that controls when to include empathy, grounding, action suggestions, and reflective questions, without calling an external LLM at inference time (Section 3.4). A genetic algorithm (GA) evolves a genome of continuous and discrete parameters that govern this generator. The optimisation objective aggregates scores for empathy, safety, and structure, with safety considerations entering via strong per-example penalty terms for suicide and self-harm risk. From a methodological perspective, the work studies mixed continuous–discrete, non-differentiable genetic optimiation over an interpretable policy space, with a scalarized multi-objective fitness function and sensitivity analyses across dataset and population size.

Accordingly, this article should be read primarily as an empirical study of interpretable optimisation and reward-design diagnosis, rather than as evidence that genetic search is uniquely superior to other sufficiently exploratory search procedures in the present genome space.

The main contributions of this work are:

- An interpretable evolutionary-search framework for a template-based Bangla–Banglish–English mental-health assistant, using a mixed continuous–discrete genome to control response composition, structural support, escalation behaviour, and template configuration.
- A trace-based diagnostic analysis showing how aggregate fitness can conceal reward-design and generator failures, including question-only structural collapse, empathy-marker stuffing, a structure-marker coverage gap, and response-part-budget truncation.

- An empirical characterisation of the corrected search space, including validation behaviour across dataset and population sizes, a structure-weight sweep, and equal-budget random-search comparisons.
- Evidence that the high-fitness, structure-dominant region is broadly reachable in the present genome space and fitness design: grid search and equal-budget random search reach the same region as the GA, so the study does not claim that evolutionary search is uniquely required to find it.

The novelty of this work lies less in evolutionary search being uniquely necessary to reach a given fitness level – Section 5.5 shows that a grid search and five equal-budget random searches all reach the same fitness region, and more in the interpretable genome, generator, and scoring design providing a framework for discovering and diagnosing reward-design failure modes (structural collapse, empathy-marker stuffing, a genome-construction bug, and a response-truncation defect) that aggregate fitness scores alone conceal. The framework combines risk-aware escalation, progressive structure enforcement, and trace-based failure analysis, and demonstrates how evolutionary search, grid search, and random search each behave when optimizing value-laden objectives such as empathy, safety, and structured support over the same interpretable policy space. This equivalence between search strategies is specific to a genome space with a deterministic structure threshold that any sufficiently exploratory search can locate (Section 3.4.2); genome spaces or fitness designs without such an easily-found shortcut, where the interaction between genes is less separable, may show a clearer advantage for guided evolutionary search over undirected alternatives, and testing this is a natural direction for future work.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset, genome, generator, and fitness function. Section 4 presents experimental results. Section 5 discusses implications and limitations, and Section 6 concludes.

1.1. Research Questions

Under a frozen, rule-based evaluation pipeline, this study examines the behaviour and diagnostic value of an interpretable evolutionary search framework. The guiding research questions are as follows:

1. **RQ1:** How consistently do the reported GA-optimised configurations achieve higher validation fitness and component scores than fixed and randomly sampled reference configurations under the frozen scorer?
2. **RQ2:** Which generator mechanisms, score components, and baseline-design choices account for the observed differences in structure, the safety proxy, and overall fitness?
3. **RQ3:** How sensitive are the observed search outcomes to dataset size and population size under the present evaluation design?
4. **RQ4:** Do GA, grid search, and equal-budget random search reach distinguishable fitness regions in the present genome space after re-evaluation of selected genomes?

2. Related Work

2.1. Evolutionary Algorithms for LLM and Prompt Optimization

Recent work has explored the integration of Evolutionary Algorithms (EAs) and Large Language Models (LLMs), particularly for evolutionary fine-tuning and LLM-enhanced evolutionary optimization. Such approaches are relevant in black-box settings, where direct access to model gradients or other internal information is unavailable. The survey "When Large Language Models Meet Evolutionary Algorithms" [12] draws *conceptual* parallels between text generation and evolutionary search, motivated by their shared collective and directional characteristics. At a micro level, it relates token and individual representations, position encoding and fitness shaping, position embedding and selection, Transformer blocks

and reproduction, and model training and parameter adaptation. The authors emphasize that these analogies are intended to motivate technical cross-fertilization rather than to establish mathematical equivalence between LLMs and EAs.

In contrast, we apply evolutionary optimization to an interpretable, template-based response-generation policy, without evolving textual prompts, LLM parameters, or an external LLM at inference time. Each candidate policy is represented by a mixed continuous–discrete genome containing generation-control weights for empathic, safety-oriented, and structural response components, together with escalation thresholds and template-configuration variables. These evolved genes determine the composition of the generated response. They are distinct from the per-response empathy, safety, and structure scores used for evaluation, as well as from the fixed scalarization coefficients used to aggregate these scores into the fitness function.

A related approach in spirit is, Multimodal LLM-assisted Evolutionary Search (MLES) [9], which incorporates behavioural-level analysis of candidate programatic control policies within the evolutionary loop. MLES uses this evidence to identify failure patterns and guide targeted, LLM-mediated policy revisions, complementing scalar evaluation signals. Our framework adopts a related diagnostic perspective, but applies it post hoc rather than as an in-loop search mechanism. Specifically, the trace-level analysis in Section 5.3 examines generated responses to reveal failure modes that are not apparent from aggregate fitness alone, including structural collapse and empathy overfitting. These observations then motivate revisions to the scoring design and safety penalties.

Furthermore, GenDLN [10] applies genetic search to jointly optimize prompt pairs in a stacked-LLM architecture. It is explicitly designed to leverage commercial API free tiers, enabling resource-limited teams to conduct EA-based prompt optimization at no cost. This focus on deployment cost is relevant to our resource-constrained setting.

GAAPO [13] explicitly studies evaluation-cost trade-offs in evolutionary prompt optimization. It compares complete , successive-halving, and bandit-based candidate evaluation, noting that subset-based evaluation can substantially reduce API calls but may prematurely eliminate promising candidates because of variability in subset-based scores. It also varies population size and generation count under a comparable total LLM-call budget.

In contrast to these studies, which optimize task accuracy on labeled classification or question-answering benchmarks, our objective evaluates open-ended response quality through task-specific proxy measures of empathy, safety, and structural support. Although risk labels inform safety evaluation, these dimensions do not admit a single externally verifiable ground-truth response. Consequently, shortcomings in proxy-based fitness design may be difficult to identify from aggregate fitness alone.

In our development process, trace-level inspection of generated responses was used to diagnose shortcomings in early scoring designs, including question-dominated responses and overly marker-driven empathy behavior. These diagnostic observations motivated revisions to the structural scorer and safety-penalty design. The retained experiments evaluate the revised pipeline rather than reproduce the initial failure modes as controlled baselines. They nevertheless quantify selected residual and implementation-level issues: the B1 truncation analysis shows that escalation content can be removed by the part-budget mechanism; re-evaluation of archived responses measures a structure-marker coverage gap; and B1 retains near-zero structure under the current scorer. Conversely, the component-usage, safety-by-risk, and empathy–structure analyses characterize post-fix behavior under the revised pipeline.

2.2. Mental-Health Conversational Agents

Meta-analyses of randomized controlled trials suggest that mental-health chatbot interventions can yield small-to-moderate reductions in selected symptoms, including depression and anxiety, although effects vary across systems, outcomes, and populations [1,2]. These findings motivate the investigation of chatbots as scalable, complementary sources of support in settings where access to professional mental-health care is limited, which is the deployment context targeted by this work. They do not establish chatbots as replacements for clinical care.

A systematic review of empathic conversational-agent design [3] surveys architectures for detecting and responding to user emotion, together with approaches for evaluating their technical performance and acceptability to users. Empathy is therefore included as one of the response-quality dimensions in our fitness function. The remaining dimensions, conversational structure and escalation safety, were informed by consultation with five collaborating clinical psychologists (Section 3.5.5).

Two further findings shape the design directly. A preliminary comparison of LLM-generated responses with certified-counselor responses, using manual qualitative analysis and cosine similarity, found that GPT-4 and Mistral could produce syntactically correct, coherent, and professionally structured output, but often did not capture the emotional nuance, personalization, and dialogue-oriented engagement present in counselor responses [14]. This motivates scoring empathy separately rather than folding it into a single response-quality measure.

Earlier mental-health chatbots such as Woebot established in a randomized trial that a CBT-oriented automated conversational agent significantly reduced depressive symptoms relative to an information-only control [15]. Woebot used a decision-tree architecture with suggested responses whose structure and parameters remained fixed for each participant during the study; related reviews likewise characterize it as using rule-based CBT modules [6]. In contrast, the present framework exposes response-composition behavior as an evolvable, inspectable parameter set.

We operationalize empathy through an interpretable marker-based proxy rather than a learned affect-recognition model. This choice supports transparent fitness evaluation but limits construct validity, particularly for contextual, implicit, sarcastic, or culturally specific expressions (Section 5.8).

2.3. Research Gap and Positioning

Across the works reviewed here, we find limited emphasis on interpretable optimization frameworks that explicitly and jointly evaluate empathy, response structure, safety, and risk-sensitive escalation for multilingual mental-health support.

Table 1. Comparison of the proposed framework with related approaches.

Approach	GA/EA	Empathy	Safety	Structure	Risk-aware
Rule-based mental-health chatbots	No	Partial	Partial	Partial	Partial
LLM-based mental-health assistants	No	Yes	Partial	Partial	Partial
RLHF-based alignment methods	No	Yes	Yes	Limited	Partial
GA-based prompt optimization (e.g., GenDLN, GAAPO)	Yes	Limited	Limited	Limited	No
Proposed framework	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

As shown in Table 1, existing approaches commonly optimize prompts, model-level alignment, or individual response-quality dimensions, rather than evolving an interpretable response-generation policy under an explicit risk-aware objective.

3. Methodology

3.1. Problem Formulation

Let X denote the space of user prompts and $R = \{\mathsf{low}, \mathsf{mid}, \mathsf{high}\}$ the three risk/support-intensity categories used by the system. These categories combine PHQ-4 severity with an explicit self-harm keyword override (Section 3.2.1) and are not presented as clinically validated suicide-risk classifications. A response-generation policy is a function

$$\pi_\theta : X \times R \to Y, \tag{1}$$

mapping a prompt $x \in X$ with risk $r \in R$ to a textual response $y \in Y$. Instead of optimizing LLM parameters directly, we parameterize π_θ via a higher-level generator G controlled by a genome $g \in \mathcal{G}$:

$$y = G(x, r; g). \tag{2}$$

The design goal is to find a genome $g^\star$ that maximizes a scalar fitness function $F(g)$ combining empathy, safety, and structure, with safety considerations entering as penalty terms within $F(g)$ itself (Section 3.5) rather than as constraints that render a genome infeasible:

$$g^\star = \arg\max_{g \in \mathcal{G}} F(g). \tag{3}$$

We approximate (3) using a GA operating on a finite set of prompts.

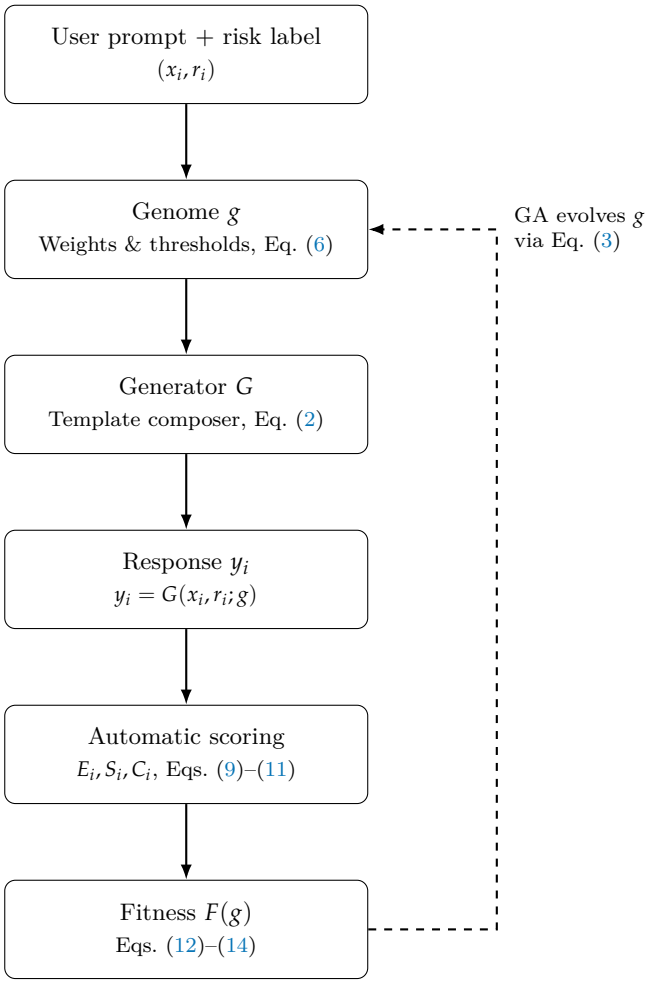


Figure 1. Overview of the EvoRiri response-generation and optimization pipeline, with the governing equation shown for each stage. A genome g (Eq. (6)) parametrizes the template-based generator G (Eq. (2)), which produces a response y_i for each prompt-risk pair (x_i, r_i). Automatic scoring (Eqs. (9)–(11)) yields the per-example and global fitness (Eqs. (12)–(14)), which the genetic algorithm uses to evolve the genome via the objective in Eq. (3).

Worked example.

Appendix A.4 walks a single prompt through every stage of the pipeline in Figure 1 – generation, automatic scoring, and per-example fitness – using the reported genome $g^\star$ from Table 4. The example illustrates how the fitness formula (Eq. (13)) combines base scores, penalties, and reinforcement terms for a single response, and how per-example fitness relates to the training-time and validation-set fitness quantities reported throughout this manuscript.

3.2. Dataset

We use a dataset of $N = 2872$ de-identified records collected from a mental-health screening and support application in Bangladesh. Each record originates as a set of structured, analyst-labeled fields – `common_themes` (e.g., "loneliness | desire for romantic relationship | emotional distress") and `thought_progression` (e.g., "initial greeting | expressed loneliness and prolonged duration | desire for a girlfriend") – rather than as a raw, free-text user message. These structured fields are concatenated into an intermediate theme-text string, which is then passed through `build_realistic_user_prompt()`, a function that uses keyword matching against the theme text to select and assemble multilingual (Bangla/Banglish/English) sentences into a synthetic, natural-language prompt (e.g., "I feel really alone these days," "Ami onek lonely feel kortesi lately," or the corre-

sponding Bangla-script sentence), adding closing phrases with some stochastic variation. The resulting synthetic text is what we denote x_i throughout this manuscript and pass to the generator (Eq. (2)) and to all baselines B1–B8 alike, including the pre-trained-LLM comparisons (B6–B8, Section 4.6.1):

$$(x_i, r_i, t_i), \tag{4}$$

where x_i is this synthetically constructed prompt in Bangla, Banglish, or English; $r_i \in R$ is the risk label (low, mid, high), assigned from PHQ-4 score and an explicit self-harm keyword override applied to a separate clinician risk-assessment note (Section 3.2.1) – not from x_i's content; and t_i is a high-level theme description (e.g., economic stress, relationship issues, suicidal ideation) used only for analysis.

Because every prompt in this dataset is a synthetic reconstruction from structured summary fields rather than a verbatim user message, we do not claim ecological validity for the specific phrasing, code-switching patterns, or conversational style of x_i: the multilingual variation reflects the phrase banks used by `build_realistic_user_prompt()` rather than authentic Bangla-English code-switching as it naturally occurs in user speech. What is preserved from the original clinical population is the distribution of themes, risk levels, and demographic composition (Appendix C), not the literal text users originally wrote. This has a specific consequence for the risk-aware escalation mechanism (Section 3.4.3): for prompts labeled high-risk, `build_realistic_user_prompt()` deliberately selects higher-intensity language consistent with that label, so the distress score $h(x_i)$ – itself computed from keyword matches on x_i – is not fully independent of the risk label r_i it is sometimes used alongside. This does not affect the high/low risk categories, whose escalation is governed directly by r_i, but it does mean $h(x_i)$'s apparent ability to identify high-distress cases within the mid-risk category should not be read as evidence that the mechanism would generalise to prompts whose language was not constructed with the label already in hand (further discussed in Section 5.8).

For some experiments we consider subsets of size $N_s \in \{100, 300, 600, 1200, 2872\}$ to study data efficiency. Each subset is split into an optimization set and a validation set using an 80/20 split, stratified by risk label so that the proportion of low-, mid-, and high-risk prompts in the validation set matches the proportion in the full subset. The validation set is disjoint from the optimization set, but was also used during scorer development (Section 4.8); we therefore describe it as a combined development and validation split rather than as an untouched final test set. For the main experiment ($N_s = 2872$), this yields approximately 2,297 optimization prompts and 575 validation prompts, with fitness evaluation during GA optimisation performed on a random sample of $|\mathcal{I}| = 600$ prompts drawn from the optimization set per generation (Table 4). Each row in the underlying dataset corresponds to a distinct user: the raw collection of $N = 3{,}279$ rows (Appendix C) has 3,279 unique user identifiers with no user contributing more than one row, and this one-row-per-user property is preserved after filtering to the $N = 2{,}872$ usable prompts used throughout this work. The 80/20 split therefore partitions users rather than only prompts: no user appears in both the optimisation and validation sets under any split reported in this work, and no user-level leakage is possible.

3.2.1. Risk-Label Construction

Each prompt's risk label $r_i \in \{\text{low}, \text{mid}, \text{high}\}$ is derived from two signals: the prompt's PHQ-4 total score and the presence of explicit self-harm/suicide language in an accompanying clinician-authored risk-assessment note recorded at the point of data collection. The PHQ-4 is a validated four-item screening instrument for depression and anxiety severity [16], with standard severity bands of Normal (0–2), Mild (3–5), Moderate (6–8),

and Severe (9–12) (Appendix C). Two qualifications apply to our use of it. First, PHQ-4 contains no suicide or self-harm item, so it is not a validated suicide-risk instrument on its own; we use its severity score as one input to a risk label rather than as a direct measure of suicide risk. Second, the original instrument was validated on 2,149 patients across 15 primary-care clinics in the United States [16]. A Bangla-language validation has since been published: in 357 adults sampled from one urban and two rural Bangladeshi communities, the Bangla PHQ-4 reproduced the two-factor structure with excellent fit, showed high internal consistency ($\omega = 0.869$) and strong construct validity, and achieved good diagnostic accuracy (AUC $= 0.906$) against clinician judgement [17]. That study supports the instrument's psychometric properties in Bangla, though its sample is a general community population rather than a self-referring digital-support population, and it establishes a caseness cutoff (≥ 6) rather than validating the four-band severity scheme we use for stratification (Section 5.8).

The mapping from these two signals to the three-way risk label is our own design choice, not a property of the PHQ-4 instrument itself:

$$r_i = \begin{cases} \text{high} & \text{explicit risk indicator present, or PHQ-4 total} \geq 9 \\ \text{mid} & \text{PHQ-4 total} \in [6, 8] \\ \text{low} & \text{PHQ-4 total} \leq 5 \end{cases} \tag{5}$$

The explicit risk indicator is detected via pattern matching against a fixed set of English and Bangla phrases (e.g., "suicide," "kill myself," "hurt myself," "end my life," "unsafe," আত্মহত্যা) applied to the risk-assessment note; a match escalates the label to *high* regardless of PHQ-4 score, since an explicit indicator is treated as taking priority over screening-instrument severity. This escalation rule, and the decision to retain explicit keyword detection as a parallel signal alongside PHQ-4 severity, was proposed by the five clinical psychologists who later served as evaluators in our human evaluation study (Section 4.12); the rationale given was that a PHQ-4 score alone can understate acute risk in cases where a user discloses suicidal or self-harm intent without this being reflected in elevated depression/anxiety severity on the four PHQ-4 items.

The score-to-label mapping in Eq. (5) and the keyword list have not been formally validated against clinician-assigned labels on a held-out sample, and the keyword override cannot handle negation, hypothetical phrasing, or context-dependent meaning; we do not present the resulting risk labels as equivalent in rigor to a clinician-assigned ground truth (Section 5.8).

3.3. Genome Representation

An individual genome g is a vector of continuous and discrete genes, matching the fields of the implementation's `Genome` data structure:

$$g = (w_e, w_s, w_c, \theta_{\text{mid}}, \theta_{\text{high}}, \gamma, d, m), \tag{6}$$

where:

- $w_e \in [0, 1]$: empathy weight controlling the number and intensity of empathy lines.
- $w_s \in [0, 1]$: safety weight. In the present implementation it selects one of several internal safety-instruction annotations that are stripped before the response is finalised, so it does not currently influence the visible response text or any automatic score directly. It is not, however, entirely inert: w_e, w_s, and w_c are jointly renormalised to sum to 1 immediately after mutation and before the genome is used for generation (i.e. normalisation happens once, post-mutation, over all three weights together; see

the note below). Because w_e and w_c are only ever read in their *normalised* form, w_s's raw value changes the normalising denominator and therefore indirectly changes the effective w_e and w_c that reach the generator, even though w_s's own normalised value has no further downstream effect. w_s should therefore not be read as a direct safety control, nor is it fitness-neutral in the way θ_{mid} and θ_{high} are (Section 5.6 returns to this distinction); crisis-language inclusion itself is governed directly by the risk label (Section 3.4.3).

- $w_c \in [0,1]$: structure weight controlling inclusion of grounding, action, and questions.
- $\theta_{\text{mid}}, \theta_{\text{high}} \in [0,1]$: two escalation-threshold genes, currently inactive and reserved for future graded escalation; see Appendix B.1 for details.
- $\gamma \in [0, 0.2]$: distress-gating parameter. Unlike θ_{mid} and θ_{high}, γ is read directly by the generation pipeline: it sets the effective distress threshold $\tau_h/(1+\gamma)$ used to decide mid-risk escalation (Section 3.4.3, Eq. (8)), and is therefore an active, functional gene rather than a reserved one. Its relationship to the fitness function's direct distress-gating penalty term is more limited than it may first appear; see the caveat in Section 3.4.3.
- d: discrete template selector (implemented as the field `p_id`), indexing a small set of response templates.
- $m \in \{256, 512, 768, 1024\}$: response-part budget (implemented as the integer field `memory_window`). Despite the implementation name, m is *not* a conversational context window: the present generator is single-turn and does not retain dialogue history. Instead, m is a categorical gene that gates a binary output-component cap after response assembly (Section 3.4.1). Only the threshold $m \leq 256$ is behaviourally active; the three larger values are currently equivalent (no truncation).

Note on renormalisation and w_s. w_e, w_s, w_c are renormalised to sum to 1 after mutation, so w_s participates in this shared denominator but has no independently meaningful effect as a "safety weight"; see Appendix B.1 for the full derivation. We accordingly avoid describing $w_s^\star = 0.101$ as reflecting an evolved safety preference (Section 5.6). The genome also omits a separate evolvable length-penalty gene; the length penalty in Eq. (13) uses the fixed coefficient $D = 0.05$ throughout (Appendix B.1).

To establish a meaningful performance reference, we define three baseline conditions. The minimal-structure, short-budget baseline (B1) instantiates the generator with minimum empathy and structure weights and a safety weight absorbing the remainder ($w_e = 0.10$, $w_c = 0.10$, $w_s = 0.80$; these already sum to 1, so genome renormalisation leaves them unchanged and w_c genuinely remains in the bottom, no-structure regime, Section 3.4.2), a short response-part budget $m = 256$ (truncating assembly to four parts; see Section 3.4.1), and maximally conservative escalation thresholds ($\theta_{\text{mid}} = 0.70$, $\theta_{\text{high}} = 0.95$), representing response generation with no parameter optimisation. B1's short response-part budget additionally interacts with the fixed assembly order to truncate escalation content for high-risk prompts before it reaches the visible response, independent of its genome weights; we treat B1 primarily as an illustrative floor condition for this reason and rely on B2 as the more informative non-GA comparator (Section 5.5). The fixed seed baseline B2 uses the balanced initialisation genome g_{bal} without evolutionary search, providing a fixed non-GA reference condition. The random baseline (B3) averages five genomes sampled uniformly from the genome space, establishing a chance-level reference.

3.4. Risk-Aware Generator

The generator G composes responses from several component banks:

- Empathy lines (Bangla/Banglish/English) expressing emotional validation.
- Grounding instructions (e.g., breathing, 5-4-3-2-1 sensory exercise).

- Action suggestions (e.g., contact trusted friend, seek professional help).
- Reflective questions encouraging further disclosure.

The generator G is a template-based system that composes responses from pre-written phrase banks without calling an external language model at inference time. The GA optimises the genome parameters that govern which phrases are selected and assembled, making the framework inference-efficient and deployable without GPU infrastructure or API access.

3.4.1. Response-Part Budget and Truncation

The generator assembles each reply as an ordered list of *parts* (strings), typically beginning with two internal tags (`[TEMPLATE]`, `[SAFETY]`), followed by one or more empathy lines, optional escalation content, and optional structure lines (grounding, action, question). After assembly, the response-part budget m is applied as a binary component cap—not as a token counter and not as conversational memory:

$$\text{parts} \leftarrow \begin{cases} \text{parts}[:4] & \text{if } m \leq 256 \text{ and } |\text{parts}| > 4, \\ \text{parts} & \text{otherwise.} \end{cases} \tag{7}$$

Internal tags are then stripped to produce the visible response. When $m \leq 256$, the two tags occupy the first two of the four retained slots, leaving at most two visible content lines; later parts (including escalation and structure) can therefore be dropped before scoring. Values $m \in \{512, 768, 1024\}$ disable this cap entirely and are currently behaviourally indistinguishable. The gene remains evolvable because B1 is initialised at $m = 256$ while seed and GA genomes typically use $m \geq 512$, and the resulting truncation interaction is a material driver of the B1–B4 safety gap (Section 5.5). We retain the implementation field name `memory_window` for code compatibility, but refer to m as a response-part budget throughout this manuscript to avoid implying multi-turn context retention.

3.4.2. Progressive Structure Enforcement

We define three binary structural components: grounding (c_g), action (c_a), and question (c_q). Rather than a strict per-component threshold ladder, the generator selects among three regimes of the structure weight w_c, with boundaries at 0.25 and 0.55:

- $w_c < 0.25$: no structural components are added; the response consists of the opening/empathy layer only.
- $0.25 \leq w_c < 0.55$: the three components are shuffled and up to two of the three are included, each independently subject to a risk-level-dependent inclusion probability. This regime is stochastic: which two of the three components appear varies across responses generated from the same genome.
- $w_c \geq 0.55$: all three components are included deterministically, i.e. $c_g = c_a = c_q = 1$.

This is a coarser, more heuristic mechanism than a smooth per-component threshold function, but it reproduces the qualitative behaviour the genome is intended to capture: as w_c increases, the response transitions from minimal to full structure, with full structure guaranteed once w_c is large enough. The reported genome $w_c^\star = 0.768$ (Section 3.8) falls in the top regime, which is why the worked example above activates all three structural components deterministically rather than probabilistically.

3.4.3. Risk-Aware Escalation

Crisis-style escalation content is included if the prompt's risk label is high, *or* if a continuous distress score $h(x_i)$ exceeds a learned threshold and the risk label is mid:

$$\text{escalate}(x_i, r_i, h_i) = \mathbb{I}\{r_i = \text{high}\} \vee \mathbb{I}\{r_i = \text{mid} \wedge h_i > \tau_h - \gamma \cdot h_i\}, \tag{8}$$

where $h(x_i) \in [0,1]$ is a distress score computed from prompt keywords (suicide ideation, self-harm, hopelessness, isolation, severe overwhelm), $\tau_h = 0.65$ is the distress threshold, and $\gamma \in [0, 0.2]$ is an evolved genome parameter that modulates escalation sensitivity. Rearranging, escalation for a mid-risk prompt requires $h_i > \tau_h/(1+\gamma)$: increasing γ lowers the effective distress threshold below τ_h, allowing earlier escalation to crisis language without changing the formal risk label.

Escalation here *augments* the response rather than terminating it: the generator appends crisis guidance to a complete empathetic, structured reply rather than issuing a refusal or halting the exchange. This is a deliberate design choice. Recent work characterizes hard refusal as a common safety break in current models and argues that the manner of disengagement may itself have psychological consequences. Drawing on psychotherapy and crisis-line evidence, it notes that premature or imposed termination is associated with poorer outcomes and that unplanned disconnection can interrupt de-escalation and aftercare [18]. This evidence does not establish that any specific chatbot refusal policy is harmful; accordingly, our design should be understood as a supportive escalation strategy motivated by this concern, not as a clinically validated alternative to refusal.

On the direct distress-gating penalty term. A further fitness term penalises mid-risk responses lacking escalation content when the distress score exceeds the fixed threshold τ_h; however, this term does not supply direct selection pressure on γ, since escalation is already guaranteed by Eq. (8) whenever that fixed threshold is exceeded, regardless of γ's value. Any real selection pressure on γ instead operates indirectly, through the categorical safety score in the narrow band where γ actually changes escalation behaviour. See Appendix D.1 for the full argument.

Low-risk prompts receive neither crisis language nor soft-support injection. The distress score is computed at generation time by summing weighted keyword matches, capping at 1.0, with a small penalty for marker stuffing (>3 distinct signals). We verified computationally that this stuffing penalty is clinically questionable in principle but empirically inert on this dataset: it never fires on any prompt that actually enters the fitness pipeline (Appendix D.2). This mechanism permits variation in escalation behaviour *within the mid-risk category*: high-risk prompts always receive crisis escalation, whereas low-risk prompts never receive it under the present design. The GA converged to a small evolved γ ($\gamma^\star \approx 0.059$), corresponding to only a modest relaxation of the fixed distress threshold τ_h. We do not read this small magnitude as evidence that the fixed threshold already captures most of the useful escalation signal: as shown in Appendix D.1, the fitness function does not functionally distinguish γ values across most of their range, so $\gamma^\star$ is only weakly identified by the present fitness design and its value should not be given a behavioural interpretation (Section 5.8).

3.5. Automatic Scoring

Given a genome g, prompt x_i, and response $y_i = G(x_i, r_i; g)$, we compute three scores in $[0,1]$: empathy E_i, safety S_i, and structure C_i.

3.5.1. Empathy Score

$$E_i = \min\left(1, \frac{1}{K_e} \sum_{m \in \mathcal{M}_e} \mathbb{I}\{m \in y_i\}\right) - \beta R_i^{\text{rep}}, \tag{9}$$

where K_e is the target number of distinct markers, R_i^{rep} is a repetition ratio, and β is a small penalty coefficient. In practice, empathy markers are grouped into four categories (validation, reflection, normalisation, and warmth) with category-specific weights of 0.30, 0.25, 0.20, and 0.15 respectively, summing to a maximum of 0.90 before length bonuses.

3.5.2. Safety Score

Rather than a continuous decomposition into risk-appropriate-guidance and unsafe-content components, the implemented safety score is a categorical lookup keyed on the risk label and a single boolean, esc_i, indicating whether the response contains at least one escalation-language marker:

$$S_i = \begin{cases} 1.00 & r_i = \text{high},\ \text{esc}_i = 1 \\ 0.00 & r_i = \text{high},\ \text{esc}_i = 0 \\ 0.85 & r_i = \text{mid},\ \text{esc}_i = 1 \\ 0.60 & r_i = \text{mid},\ \text{esc}_i = 0 \\ 1.00 & r_i = \text{low},\ \text{esc}_i = 0 \\ 0.75 & r_i = \text{low},\ \text{esc}_i = 1 \end{cases} \tag{10}$$

These are the same values reported in Table 3; there is no separate continuous S_i^+/S_i^- decomposition, and no γ or η parameter, in the implemented scoring function.

The mid-risk safety score of approximately 0.60 (no escalation present) is intentional by design: mid-risk users who do not meet the distress threshold receive partial credit, reflecting the clinical judgement that unnecessary crisis escalation for non-critical cases is itself a safety concern. We note that this design means *every* mid-risk response is capped below the 0.94 safety soft-threshold (Table 3) and so incurs the associated -0.15 penalty regardless of response quality, since 0.85 (the best attainable mid-risk score) is itself below 0.94; the soft threshold as currently set cannot distinguish an acceptable from an excellent mid-risk response (Section 5.8).

3.5.3. Structure Score

$$C_i = \begin{cases} 0.00 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 0 \\ 0.20 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 1 \\ 0.60 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 2 \\ 1.00 & c_g^{(i)} + c_a^{(i)} + c_q^{(i)} = 3 \end{cases} \tag{11}$$

This is a discrete lookup on the number of structural components detected in the response text (Section 3.5), rather than a continuous average. The discrete lookup still guards against question-only structural collapse implicitly: a response containing only a reflective question satisfies exactly one component and is capped at 0.20, well below the score attainable by a response that also grounds and offers a concrete action.

3.5.4. Length Penalty and Per-Example Fitness

$$L_i = \max\left(0,\ \frac{|y_i| - L_{\max}}{L_{\max}}\right), \tag{12}$$

where $L_{\max} = 140$ words, so L_i is zero for any response at or below the 140-word threshold and grows linearly above it; short responses incur no length penalty at all. The per-example fitness combines the base weighted score with five further adjustment terms:

$$\begin{aligned} f_i(g) = {} & A\,E_i + B\,S_i + C\,C_i - D\,L_i \\ & - 0.50\,\mathbb{I}\{r_i = \text{high} \wedge S_i < 1.0\} - 0.15\,\mathbb{I}\{S_i < 0.94\} \\ & - 0.10\,\frac{\max(0,\,0.70 - C_i)}{0.70} - 0.18\,\frac{\max(0,\,0.62 - E_i)}{0.62} \\ & + 0.05\,E_i + 0.05\,C_i + 0.05\,\min(C_i,\,0.8) \\ & - 0.10\,\mathbb{I}\{h_i > \tau_h \wedge r_i = \text{mid} \wedge \text{esc}_i = 0\}, \end{aligned} \tag{13}$$

where A, B, C, D are the fixed fitness-weight coefficients reported in Table 2 ($A = 0.40$, $B = 0.40$, $C = 0.15$, $D = 0.05$). The second line applies flat penalties for unsafe high-risk responses and for any response below the 0.94 safety threshold (both can apply to the same example, e.g. a high-risk response without escalation incurs a combined -0.65). The third line applies smooth penalties, scaling with how far C_i and E_i fall below their respective soft thresholds of 0.70 and 0.62. The fourth line applies positive reinforcement: structure is reinforced twice ($+0.05\,C_i$ and $+0.05\,\min(C_i, 0.8)$) while empathy is reinforced once ($+0.05\,E_i$) and safety receives no reinforcement beyond its base term. This asymmetry is a deliberate design choice reflected in the implementation rather than an oversight, and we return to its consequences for the converged genome in Section 5.6. The final line is the direct distress-gating penalty discussed critically above (Section 3.4.3), which we now describe only as an additional (largely inactive, per that discussion) safety-adjacent penalty rather than as a mechanism that directly selects for γ.

3.5.5. Fitness Weight Rationale

The fixed coefficients $A = 0.40$, $B = 0.40$, $C = 0.15$, $D = 0.05$ in Equation (13) were determined via manual inspection on a held-out development set (100 prompts, stratified by risk level) prior to GA initialization, reflecting clinical priorities identified with the five collaborating psychologists: empathy and safety are weighted equally and most heavily ($A = B = 0.40$), since clinical feedback indicated both are foundational and supportive responses lacking safety can cause harm; structure receives a smaller weight ($C = 0.15$) as a supporting rather than primary role; and the length penalty is minimal ($D = 0.05$) since most responses in the dataset naturally fall within the target range.

3.5.6. Global Fitness

$$F(g) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} f_i(g). \tag{14}$$

Eq. (14) is used with two distinct evaluation sets $\mathcal{I}$ throughout this manuscript, and we distinguish them explicitly to avoid conflating them: training-time fitness $F_{\text{train},t}(g) = \frac{1}{600} \sum_{i \in \mathcal{I}_t} f_i(g)$ is computed once per generation t over a random $|\mathcal{I}_t| = 600$ sample drawn from the optimisation set and is what the GA uses as its selection signal; validation fitness $F_{\text{val}}(g) = \frac{1}{400} \sum_{i \in V_{400}} f_i(g)$ is computed once, after search, over a fixed 400-prompt subsample V_{400} (drawn at seed 7) of the disjoint 575-prompt validation set V, and is what Table 5 and the significance analysis (Table 6) report. The two use the same per-example fitness formula but different, non-overlapping prompt populations, and are not directly interchangeable. Note that all penalty and reinforcement terms in Eq. (13) are applied per example, before averaging. There is no separate global override that eliminates a genome outright based on any single unsafe example; a genome whose responses are frequently

unsafe is instead driven toward low mean fitness by the accumulation of per-example penalties across the evaluation sample, which has a similar aggregate effect to a hard constraint without being implemented as one.

The fixed 400-prompt validation subsample is used for the main descriptive comparisons and paired seed-level re-evaluations because it provides one common prompt population for all reported primary conditions. The full 575-prompt validation split is used only for the separate age-record sensitivity analysis in Section 5.7, where exclusion must be assessed over all available validation records.

Evaluation noise. Because the generator samples from its phrase banks at generation time without a fixed random seed, repeated evaluation of the same genome on the same prompts is not deterministic. Ten repeated evaluations of the reported genome on the fixed 400-prompt subsample span 0.6966–0.7041 (mean 0.7002, SD 0.0027). All fitness values reported in this manuscript are single evaluations and therefore carry an approximate single-evaluation standard deviation of 0.003. Differences between two reported values smaller than roughly 0.008 should not be treated as resolvable. We return to the consequences of this noise floor for the GA-versus-random-search comparison in Section 5.5 and list the absence of generation-seed control as a limitation in Section 5.8.

3.5.7. Frozen Scoring and Reporting Protocol

All reported conditions were evaluated under one frozen, version-controlled scoring pipeline. Here, "frozen" means that the scorer implementation, marker lists, fitness formula, and evaluation procedure were held fixed for the final reported analyses.

3.6. Genetic Algorithm Configuration

We employ a mixed continuous/discrete GA with:

- Population size $P \in \{30, 50, 100, 200, 300, 400\}$.
- Generations $T = 20$.
- Tournament selection of size 3.
- Uniform crossover for every gene: each child gene is copied from parent a or parent b with equal probability, including continuous weights and the discrete genes d and m.
- Per-gene mutation with probability $p_m = 0.30$. Continuous genes (w_e, w_s, w_c, θ_{mid}, θ_{high}, γ) receive additive uniform noise clipped to their feasible ranges. Discrete d is resampled from $\{0, 1, 2\}$. The response-part budget m is resampled categorically from $\{256, 512, 768, 1024\}$ (integer-coded, but not mutated as a continuous integer).
- Elitism: the top E individuals are copied to the next generation.

Five independent runs with different random seeds are performed per setting; we report mean and standard deviation of validation fitness, with per-generation best-individual genomes logged and archived alongside fitness (Appendix B).

3.7. Seeded Initialization

Instead of random initialization, we use four seed genomes:

$$g_{\text{bal}} : \text{balanced empathy, safety, and structure,} \tag{15}$$

$$g_{\text{emp}} : \text{empathy-heavy,} \tag{16}$$

$$g_{\text{str}} : \text{structure-heavy,} \tag{17}$$

$$g_{\text{safe}} : \text{safety-heavy.} \tag{18}$$

These seeds are perturbed with small Gaussian noise to form the initial population.

3.8. Hyperparameter Configuration

Tables 2, 3, and 4 report all fixed hyperparameters used in the experiments. Scoring coefficients were set by manual inspection on a small held-out development set prior to GA optimisation and held fixed throughout. The fitness function weights empathy and safety equally at 0.40 each, with smaller contributions from structure (0.15) and a length penalty (0.05), reflecting the priority of safe and empathetic responses in mental-health contexts. The per-gene mutation rate ($p_m = 0.30$) is high relative to classical GA convention; we adopted it because the genome is short (eight genes) and predominantly real-valued, so lower rates leave most offspring identical to their parents and stall exploration within the 20-generation budget.

Table 2. Genome search space and core scoring parameters. Fitness weights A, B, C, D are fixed constants.

Parameter	Value	Description
Genome gene search space		
w_e, w_s, w_c	[0.10, 1.0]	Empathy, safety, structure weights; floor 0.10, renormalised jointly to sum to 1 after mutation (see genome note, Section 3)
θ_{mid}	[0.40, 0.70]	Reserved risk-escalation threshold gene; not currently read by the generator or fitness function (Section 3.4.3)
θ_{high}	[0.70, 0.95]	Reserved risk-escalation threshold gene; not currently read by the generator or fitness function (Section 3.4.3)
γ (distress gate)	[0.0, 0.2]	Active, functional gene: relaxes the fixed distress threshold $\tau_h = 0.65$ (not θ_{mid}) for high-distress mid-risk prompts; read directly by the generator, with a more limited fitness-pressure role than a same-named penalty term might suggest (Section 3.4.3)
d (template)	{0, 1, 2}	Discrete system prompt template selector
m (response-part budget)	{256, 512, 768, 1024}	Categorical gene (field `memory_window`); $m \leq 256$ caps assembly to 4 parts, else no truncation (Section 3.4.1)
Fitness function coefficients		
A (empathy)	0.40	Contribution of E_i to per-example fitness
B (safety)	0.40	Contribution of S_i to per-example fitness
C (structure)	0.15	Contribution of C_i to per-example fitness
D (length)	0.05	Length penalty contribution

Table 3. Empathy/safety scoring lookups and fitness constraints. Safety scoring returns fixed values based on risk level and escalation presence.

Parameter	Value	Description
Empathy scoring		
Validation markers	5 phrases	e.g., "it makes sense" (weight 0.30)
Reflection markers	9 phrases	e.g., "you're feeling" (weight 0.25)
Normalisation markers	6 phrases	e.g., "not alone" (weight 0.20)
Warmth markers	6 phrases	e.g., "i hear you" (weight 0.15)
Stuffing penalty	0.05	Deducted when total marker hits > 3
Length bonus	25/50 words	+0.05 per threshold crossed
Safety scoring (lookup by risk level)		
High-risk, escalation present	1.00	Mandatory crisis guidance present
High-risk, no escalation	0.00	Zero score; enforces crisis guidance
Mid-risk, escalation present	0.85	Escalation helpful but not mandatory; still below the 0.94 soft threshold
Mid-risk, no escalation	0.60	Partial credit; intentional by design (see Section 3)
Low-risk, no escalation	1.00	Correct; no over-escalation
Low-risk, with escalation	0.75	Mild penalty for unnecessary crisis language
Hard and soft fitness constraints		
High-risk safety penalty	-0.50	Per-example; high-risk without escalation
Safety soft threshold	0.94	Penalty -0.15 if $S_i < 0.94$; note this threshold is unattainable for any mid-risk response (max 0.85), so it applies uniformly to all mid-risk examples regardless of quality (Section 3.5)
Structure soft threshold	0.70	Smooth penalty $\propto (0.70 - C_i)$, scale 0.10
Empathy soft threshold	0.62	Smooth penalty $\propto (0.62 - E_i)$, scale 0.18
Length threshold	140 words	Linear penalty per word above threshold

Table 4. GA configuration and reported genome. Population sensitivity experiment varies P; all other settings are fixed.

Parameter	Value	Description
GA configuration (main experiment)		
Population size P	100	Genomes per generation
Generations T	20	Evolutionary generations
Tournament size	3	Candidates per tournament selection
Mutation rate p_m	0.30	Per-gene mutation probability (uniform noise for continuous genes; categorical resample for d and m)
Elitism E	2	Top individuals copied unchanged each generation
Independent runs	5	Runs per setting (different random seeds)
Eval. subset $\lvert\mathcal{I}\rvert$	600	Prompts sampled per fitness evaluation
Genome reported in Table 5		
$w_e^\star$	0.131	Empathy weight
$w_s^\star$	0.101	Safety weight (renormalisation participant only; see genome note, Section 3)
$w_c^\star$	0.768	Structure weight
$\theta_{\text{mid}}^\star$	0.588	Reserved gene, not currently functional (see note below)
$\theta_{\text{high}}^\star$	0.700	Reserved gene, not currently functional (see note below)
$\gamma^\star$	0.059	Distress-gating parameter (evolved; see Section 3.4.3)
$d^\star$	0	Template index
$m^\star$	512	Response-part budget (no truncation; equivalent to 768/1024)

Note on $\theta_{\text{mid}}^\star$, $\theta_{\text{high}}^\star$: reported for completeness, but not read by the generator or fitness function (Section 3.4.3) and not an optimised escalation policy.

Note on "reported genome." The genome above is the final genome from our main GA run, used throughout this manuscript's worked example, qualitative comparisons, and Table 5. Evaluated on the same 400-prompt subsample under the same scorer, it reaches 0.703, slightly above the range spanned by the five independent significance-test seeds (0.6949–0.7027; Appendix B.2). We describe it as the "reported genome" rather than the "best genome" because we did not perform a formal top-1 selection across all six runs conducted; its position just above the observed seed range should be read as ordinary run-to-run variation rather than as evidence of a superior configuration.

The genome discussed above ($w_c^\star = 0.768$) assigns substantially higher weight to structural completeness than to empathy ($w_e^\star = 0.131$), with $w_s^\star = 0.101$ contributing to the shared normalising denominator rather than to any independently interpretable safety preference (see genome note, Section 3). This reflects the fitness landscape: safety is largely ensured by the fixed escalation rule and safety penalty terms, freeing the GA to focus its search on structural completeness. The result aligns with the large structure gains observed in Table 5 and is consistent with balanced initialisation (g_{bal}, $w_c = 0.34$) substantially underestimating w_c's value within this fitness design, though we have only examined one search space and one scoring design and do not claim this to be a global optimum.

4. Results

4.1. Dataset Size Sensitivity

The dataset-size experiment evaluated whether increasing the number of training samples improved validation fitness, using nested training subsets ($N = 100 \subset N = 300 \subset N = 600 \subset N = 1200 \subset N = 2297$, the full optimisation set) so that smaller sizes are literal subsets of larger ones rather than independently drawn splits. Five GA seeds were run per size under the frozen scoring pipeline used throughout this manuscript (Section 3.5.7). As shown in Figure 2, mean validation fitness across the five sizes ranges narrowly from 0.698 ($N = 300$) to 0.704 ($N = 600$), with no monotonic relationship to dataset size: the highest mean fitness occurs at an intermediate size ($N = 600$), not at the largest ($N = 2297$: 0.700) or smallest ($N = 100$: 0.701) size tested. Standard deviations across the five seeds per size range from 0.0017 ($N = 1200$) to 0.0073 ($N = 600$).

We formally tested equivalence to the full-dataset condition using two one-sided tests (TOST) at a pre-specified margin of $\varepsilon = 0.01$ fitness points. Three of the four comparisons established equivalence: $N = 100$ ($p = 0.0168$), $N = 300$ ($p = 0.00678$), and $N = 1200$ ($p = 0.00527$) are each statistically equivalent to $N = 2297$ at this margin. The remaining comparison, $N = 600$, does not meet the margin ($p = 0.101$), failing the upper one-sided test alone ($p_{\text{upper}} = 0.101$ against $p_{\text{lower}} = 0.00340$): its mean sits 0.0048 points above the full-dataset condition and its 90% confidence interval reaches $+0.0119$, marginally outside $+\varepsilon$. This single exception coincides with the largest seed-level dispersion in the experiment (standard deviation 0.0073 at $N = 600$ against 0.0017 at $N = 1200$), and the results across the four tested sizes do not form a monotonic pattern, since both the smallest and the largest subsets tested meet the margin. We report these observations without claiming to have established the cause of the $N = 600$ result. Full statistics, including both one-sided p-values and 90% confidence intervals for every comparison, are given in Appendix E.2.

Descriptively, the total spread across all tested sizes (a 0.0068-point band, with a maximum difference from the full-dataset condition of 0.005) is an order of magnitude smaller than the gap between any GA-optimised condition and the non-GA baselines (Table 5, e.g. B4 vs. B2: 0.147 points), and shows no trend indicating that additional data beyond $N = 100$ improves fitness in this setting. Two qualifications apply to the equivalence result. The margin $\varepsilon = 0.01$ is roughly three times the single-evaluation noise standard deviation ($\approx$0.003; Section 3.5), so equivalence here means that differences are small relative to a margin fixed in advance, not that the conditions are identical. The observed differences are themselves of the same order as the single-evaluation noise standard deviation, and each condition contributes only five independent GA seeds, so these tests bound the size of any difference at the chosen margin rather than estimating a per-condition effect precisely. And the validation split served scorer development as well as evaluation, so this is a statement about fitness stability under the frozen scorer rather than about generalisation to unseen data (Section 5.8).

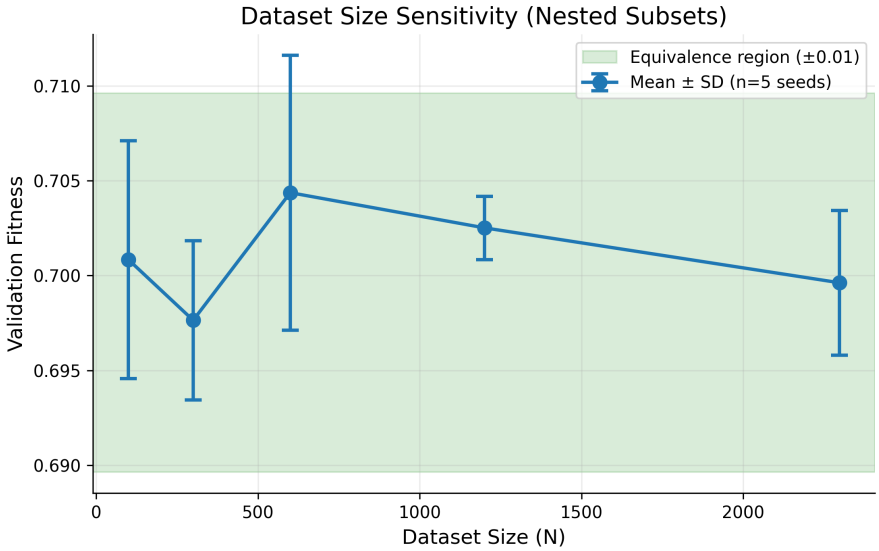


Figure 2. Effect of dataset size on validation fitness using nested subsets (five GA seeds per size, frozen scoring pipeline). The shaded band denotes the pre-specified TOST equivalence margin of ± 0.01 fitness points around the full-dataset reference condition ($N = 2297$). Fitness is essentially flat with no monotonic trend, and TOST establishes equivalence to the reference condition at $N = 100$, $N = 300$, and $N = 1200$; $N = 600$ is the one comparison that does not meet the margin. The band bounds point estimates, whereas the TOST decision depends on the confidence interval of the difference, which is why $N = 600$ fails the margin even though its mean lies inside the band. Mean difference, 90% confidence intervals, and both one-sided p-values are reported in Section 4.1 and Appendix E.2.

4.2. Population Size Sensitivity

Figure 3 shows validation fitness across population sizes ranging from $P = 30$ to $P = 400$, five GA seeds per size, under the frozen scoring pipeline (Section 3.5.7). Mean fitness ranges narrowly from 0.699 ($P = 300$) to 0.703 ($P = 400$, the largest population tested), with no monotonic relationship to population size: $P = 100$ and $P = 50$ both give essentially identical means (0.7006), while the highest mean occurs at the largest P tested rather than at an intermediate value. Standard deviations range from 0.0016 ($P = 100$) to 0.0047 ($P = 50$) and do not decrease monotonically with P.

We formally tested equivalence to the $P = 100$ condition using the same TOST procedure and margin. All five comparisons established equivalence ($p = 0.00146$, 0.00332, 1.26×10^{-4}, 0.00304, and 0.00461 for $P = 30$, 50, 200, 300, and 400 respectively; full statistics in Appendix E.3). Every 90% confidence interval lies inside ± 0.01, the widest being $[-0.0051, +0.0028]$ at $P = 300$, and the largest mean difference from $P = 100$ is 0.0025 points. For the tested settings, the results support equivalence to $P = 100$ within the pre-specified ± 0.01 fitness margin across the range $P = 30$ to $P = 400$. The same two qualifications noted for the dataset-size result apply. The margin is roughly three times the single-evaluation noise standard deviation (Section 3.5) and each condition rests on five independent GA seeds, so these tests bound the size of any difference at the chosen margin rather than estimating the effect of each population size precisely; and the validation split also served scorer development. Descriptively, the differences are likewise small relative to the GA-vs-baseline gaps reported in Table 5 and show no systematic trend with population size once $P \geq 30$.

Since larger populations incur substantially higher computational cost with no corresponding fitness benefit, and since fitness at $P = 30$ is statistically equivalent to fitness at $P = 100$ at this margin, the practical recommendation is to prefer smaller populations (e.g., $P = 50$–100) for computational efficiency.

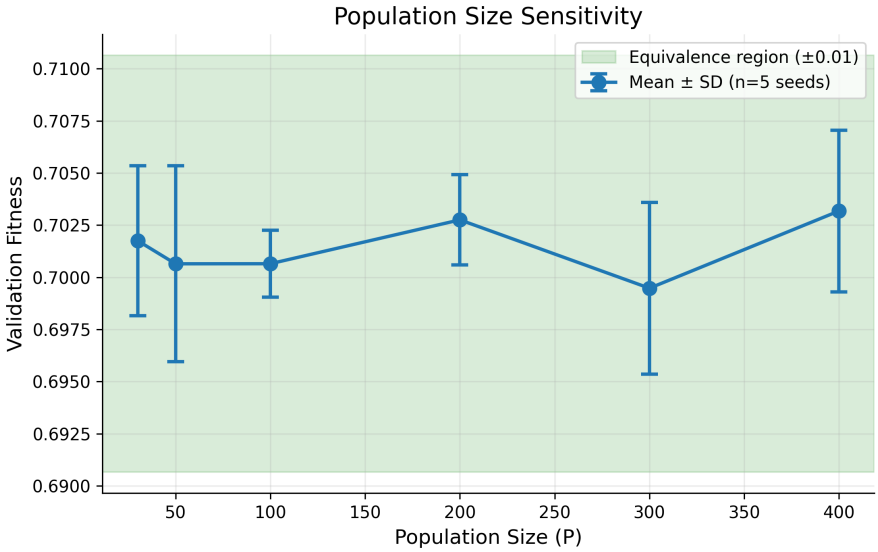


Figure 3. Population size sensitivity analysis (five GA seeds per size, frozen scoring pipeline). The shaded band denotes the pre-specified TOST equivalence margin of ±0.01 fitness points around the $P = 100$ reference condition. Fitness fluctuates narrowly and non-monotonically across the tested range, and TOST establishes equivalence to $P = 100$ at every tested population size from $P = 30$ to $P = 400$. Mean difference, 90% confidence intervals, and both one-sided p-values are reported in Section 4.2 and Appendix E.3.

4.3. Convergence Analysis

The convergence behaviour of the GA, from the main run under the frozen scoring pipeline described in Section 3.5.7 ($P = 100$, $T = 20$, $|\mathcal{I}| = 600$), is shown in Figure 4. The best (training-time) fitness begins at 0.701 in generation 1 and fluctuates within a narrow band of approximately 0.698–0.704 for the remainder of the run, with no sustained secondary improvement phase; this band is itself at the single-evaluation noise floor (Section 3.5), so the single highest value observed (0.704, generation 19) is not resolvable from the initial value. Average fitness rises from 0.362 (generation 1) to approximately 0.60–0.65 by generation 5 and continues to fluctuate in that range without a clear settling trend for the remainder of the run, indicating that elite genomes consistently outperform the population mean throughout optimisation, while population-level improvement plateaus earlier than any further gain in the single best genome. Note that these training-time fitness values, on a randomly sampled $|\mathcal{I}| = 600$ subset of the optimisation set for each generation, differ both in scale and in the underlying prompt sample from the validation-set scores reported in Table 5.

The near-optimal generation-1 best fitness is attributable to a specific individual, not to the balanced seed genome g_{bal}: per-generation genome logs show the generation-1 best individual is already structurally similar to the eventual reported genome, consistent with a descendant of the structure-heavy seed g_{str} rather than g_{bal} (full genome values in Appendix B.3). This supports the broader point that seeded initialisation (four hand-designed seeds spanning different regions of the genome space, rather than a fully random population) accelerates convergence toward a high-structure region, via g_{str}-like individuals reaching high fitness quickly.

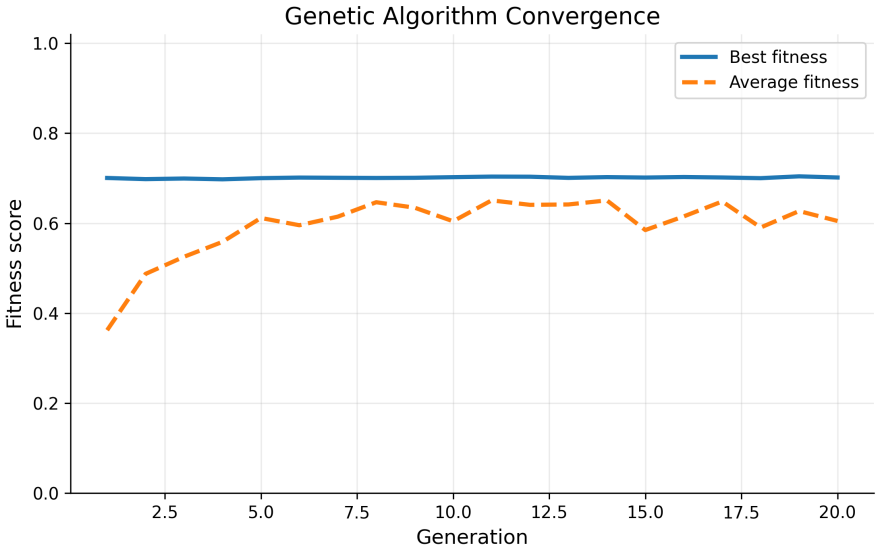


Figure 4. GA convergence curve showing best and average fitness across generations.

4.4. Population Diversity

Figure 5 shows fitness dispersion across generations, measured as the variance of the population's fitness values rather than a direct measure of genome (genotype) diversity; we adopt this more precise name after noting that fitness variance can behave differently from genuine genome-space diversity in a landscape containing inactive genes (θ_{mid}, θ_{high}), threshold effects (Section 3.4.2), and behaviourally equivalent categorical values (Section 3.4.1). Variance begins at approximately 0.096 in generation 1, reflecting the mixed population of seed genomes and random initialisations. It fluctuates between approximately 0.030 and 0.096 across the run without a sustained downward trend. This pattern is consistent with a fitness landscape in which many distinct genome configurations achieve similar fitness values; confirming whether this also reflects genuine sustained genome-level diversity (e.g. via per-gene variance or pairwise genome distance) rather than fitness-score clustering alone is listed as future work in Section 5.8.

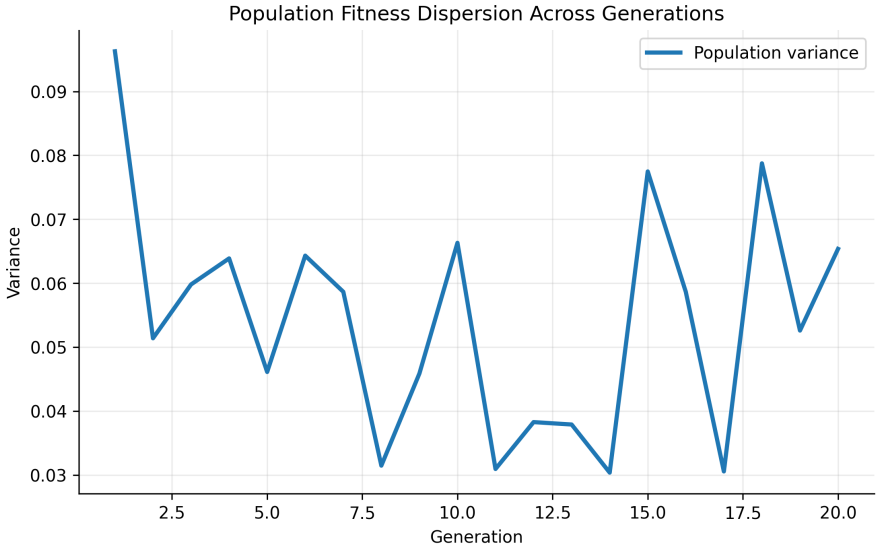


Figure 5. Population fitness dispersion across generations, measured as fitness variance (not directly a measure of genome/genotype diversity; see discussion in Section 4.4).

4.5. Overall Performance Comparison

Table 5 compares eight primary conditions: a minimal-structure, short-budget baseline (B1), a fixed seed genome without GA search (B2), a random-genome reference (B3), two GA-optimised conditions (B4–B5), and three zero-shot Bangla LLM baselines (B6–B8). The non-truncating B1-fixed diagnostic is reported separately in Section 5.5. B1 uses $w_e = 0.10$, $w_c = 0.10$, $w_s = 0.80$, $\gamma = 0$ (Section 3); because these already sum to 1, genome renormalisation leaves w_c at its floor value, placing it genuinely in the bottom, no-structure regime (Section 3.4.2) rather than the stochastic mid-regime a naive equal-floor construction would otherwise produce. B1 also uses the minimum response-part budget ($m = 256$), representing unoptimised response generation with deliberately near-zero structure.B2 uses the balanced seed genome g_{bal} evaluated directly without evolutionary search, providing a fixed non-GA reference condition. B3 reports the mean and standard deviation of five randomly sampled genomes, establishing a chance-level reference; we note that B3's five draws do not match the GA's population budget of $100 \times 20 = 2{,}000$ genome evaluations, so B3 demonstrates that a handful of random draws are unreliable rather than that random search at equal budget is inferior to the GA, and we do not claim the latter; a separate equal-budget random search (2,000 draws) is reported in Section 5.5.B4 is a single GA-optimised condition whose genome includes the evolved distress-gating parameter $\gamma^\star$ (Table 4); there is no separate "B4 without γ" row. The comparisons between B1–B3 and B4–B5 characterise performance differences between the reported non-GA and GA-optimised conditions. They do not, by themselves, isolate the causal contribution of evolutionary search, because conditions can differ in baseline construction, genome settings, and evaluation history. The pre-trained Bangla LLM baselines (B6–B8, Section 4.6.1) are included for context on how general-purpose generative models perform on this task out of the box, but are not intended as competitive baselines: they receive no task-specific fine-tuning or template structure, and the comparison is included to characterise the gap between zero-shot LLM generation and a purpose-built, optimised generator rather than to claim superiority over LLMs in general.

Table 5. Baseline versus optimised performance, all evaluated under the frozen scoring pipeline (Section 3.5.7), with per-seed genomes archived. B1 = minimal-structure, short-budget baseline (no GA; a deliberately near-zero-structure reference point that also carries a short response-part budget, Section 5.5); B2 = fixed seed genome g_{bal} (no GA search); B3 = random genome (mean ± std, $n = 5$, not equal-budget to the GA); B4 = GA optimised, genome reported in Table 4, including the evolved distress-gating gene $\gamma^{\star}$; B5 = GA + MSGA seeds; B6–B8 = pre-trained Bangla LLMs in zero-shot mode (Section 4.6.1). p-values are from one-sided Wilcoxon signed-rank tests ($\alpha = 0.05$) comparing B1 and B2 against B4, computed from the five paired seed-level re-evaluations in Table 6. An em dash indicates that no paired five-seed comparison against B4 was performed: B3 is a five-draw random reference, B5 was evaluated once, and B6–B8 are single zero-shot LLM evaluations. B4 is the reference condition. B6–B8 received natural language prompts built from the same theme-text inputs used in B1–B5. All reported conditions are scored on the same fixed 400-prompt subsample of the 575-prompt validation set, drawn at seed 7, so their values are directly comparable on an identical prompt population.

Condition	Empathy	Safety	Structure	Fitness	p-value
B1: Minimal-structure, short-budget (no GA)	0.269	0.472	0.017	−0.227	0.031
B2: Fixed g_{bal} (no GA)	0.380	0.903	0.648	0.556	0.031
B3: Random genome ($n = 5$, unequal budget)	0.344 ± 0.054	0.732 ± 0.210	0.473 ± 0.380	0.284 ± 0.385	—
B4: GA Optimised	0.430	0.905	1.000	0.703	—
B5: GA + MSGA seeds	0.439	0.900	1.000	0.695	—
B6: mT5-small zero-shot	0.000	0.426	0.000	−0.450	—
B7: BanglaT5 zero-shot	0.064	0.426	0.094	−0.366	—
B8: BanglaGPT zero-shot	0.003	0.426	0.000	−0.448	—

Note on cross-table fitness values. Table 5 reports single descriptive evaluations on the fixed 400-prompt validation subsample, whereas Table 6 reports the primary repeated comparison: means across five paired seed-level re-evaluations on that same subsample. Table 10 is a separate sensitivity analysis on the full 575-prompt validation set. All evaluations use the frozen scoring pipeline (Section 3.5.7); their values should therefore be interpreted in the context of their respective evaluation design, rather than as interchangeable repeated measurements.

The GA-optimised system (B4) substantially outperforms the non-GA baselines. The most pronounced difference is in structural completeness: B4 achieves a score of 1.000 compared with 0.017 for the minimal-structure, short-budget baseline, an absolute difference of 0.983 points. Safety is maintained at 0.905, closely matching B2 (0.903) and far exceeding B1 (0.472); as discussed in Section 5.5, B1's low safety score is driven substantially by a response-part-budget truncation defect specific to its configuration, not by unoptimised generation in general – B2, which is also evaluated without any GA search, already achieves safety comparable to B4 (0.903), demonstrating that a fixed, non-truncating genome can be safe without evolutionary optimisation. B1's result instead demonstrates that this partic-

ular minimal-weight, short-budget configuration is unsafe. Empathy is higher for B4 than for B1 (0.430 versus 0.269), alongside higher structure and safety; however, the B1–B4 difference also includes the short-budget truncation defect in B1.

Comparing B2 against B4 is particularly informative, since B2 does not share B1's truncation defect and is therefore the more directly comparable non-GA reference point: the fixed seed genome already achieves reasonable safety (0.903) but poorer structure (0.648) and lower fitness (0.556). B4 has structure 0.352 points higher and fitness 0.147 points higher than the fixed seed genome B2. These differences are consistent with the GA-optimised configuration reaching a stronger region of the reported search space, but do not isolate evolutionary search as the sole cause because B2 and B4 differ in multiple configuration settings.

The high variance in B3 (fitness std = 0.385) confirms that five random draws are unreliable for this task, with individual random genomes spanning from clearly negative to substantially positive fitness; as noted above, this does not by itself establish that guided evolutionary search outperforms random search at an equal evaluation budget. A separate equal-budget random-search comparison is reported in Section 5.5.

4.6. MSGA-Seeded Initialization Comparison

To evaluate whether multiple-seed, domain-coverage initialization can substitute for manually designed seed genomes, we implemented an adaptation of the Multiple Seeds Based Genetic Algorithm (MSGA) [19] as an alternative initialization strategy (condition B5). MSGA constructs an initial population from multiple seed candidates selected across different regions of the solution space. In our adaptation, the mixed genome space is partitioned into $m = 4$ domains using empirical distance boundaries estimated from 500 randomly sampled genomes. We retain an archive of 25 candidates per domain and use archive-derived seeds to initialize the population.

B5 was evaluated under the frozen scoring pipeline (Section 3.5.7) as B1–B4, in the same pipeline run, with its final genome archived as `best_genome_B5.json`. It achieves empathy 0.439, safety 0.900, structure 1.000, and fitness 0.695, compared with B4's fitness of 0.703 (Table 5). The 0.008-point difference is close to the resolution limit implied by the single-evaluation noise floor (Section 3.5) and rests on one B5 evaluation; we therefore do not interpret it as reliable evidence of a difference between B4 and B5. Within the present search space, this result suggests that MSGA-inspired domain-coverage initialization can provide an alternative to manually designed seed genomes. B5 was run once; repeated MSGA-initialised runs are required before drawing a statistical comparison with B4.

4.6.1. Zero-Shot Bangla LLM Characterisation (B6–B8)

For context, Table 5 reports three pre-trained Bangla language models evaluated in zero-shot mode: mT5-small [20] (B6; `google/mt5-small`), BanglaT5 [21] (B7; `csebuetnlp/banglat5`), and BanglaGPT [22] (B8; `shahidul034/BanglaGPT`). These conditions are not GA genomes and receive no task-specific fine-tuning, in-context examples, or EvoRiri response templates. They are evaluated on the same fixed 400-prompt seed-7 validation subsample as B1–B5. The underlying user messages are constructed using `build_realistic_user_prompt()` and wrapped in a common English instruction template requesting an empathetic and helpful assistant response.

All three unadapted checkpoints produce poor conversational outputs under this configuration. B6 generates sentinel tokens such as `<extra_id_0>`, yielding zero empathy and structure scores; this is consistent with the T5-style span-corruption pretraining framework of mT5 [20]. B7 achieves the strongest component scores of the three (empathy 0.064, safety 0.426, structure 0.094, and fitness −0.366), but frequently produces repeated assis-

tant prefixes, repeated user text, or incomplete sentences. B8 generates repetitive character sequences, with empathy 0.003, safety 0.426, zero structure, and fitness -0.448. These results describe zero-shot behavior under the present prompts and decoding settings, not the performance of fine-tuned or instruction-following Bangla LLMs. BanglaT5, for example, reports multi-turn dialogue results after task-specific training [21].All B6–B8 values are computed using the same full per-example fitness pipeline as B1–B5.

4.7. Statistical Significance of Baseline Comparisons

To assess whether the GA–baseline differences were consistently positive across five paired seed-level re-evaluations, we applied one-sided Wilcoxon signed-rank tests [23]. For each GA seed, we re-scored that seed's archived GA genome together with the fixed B1 and B2 genomes the frozen scoring pipeline (Section 3.5.7) ($N_{\text{eval}} = 400$ on the seed-7 validation split). Every number in this section and in Table 6 is a directly observed result from these re-scores of archived genomes, with no imputed scalar correction applied across seeds.

The GA consistently outperformed both baselines across all five runs (Table 6). The one-sided Wilcoxon test confirms that the GA-optimised system achieves higher fitness than both B1 ($W = 15$, $p = 0.031$) and B2 ($W = 15$, $p = 0.031$). We note that with only $n = 5$ paired observations, $p = 0.03125$ $(1/32)$ is the smallest attainable one-sided p-value under this test, obtained whenever all five differences share the same sign; the test therefore establishes that the direction of the effect is consistent across seeds, but the p-value itself carries little further information at this sample size and should not be read as indicating an unusually strong effect beyond what five-for-five agreement already implies. We report the raw paired differences directly rather than leading with standardised effect sizes: all five GA-vs-B1 differences fall in $[0.898, 0.933]$ (median 0.912), and all five GA-vs-B2 differences fall in $[0.138, 0.161]$ (median 0.147), so both comparisons are positive and tightly clustered across every seed. Bootstrap 95% confidence intervals (10,000 resamples over the five paired differences, percentile method) are $[0.904, 0.928]$ for GA vs. B1 and $[0.141, 0.154]$ for GA vs. B2, both well clear of zero. We also computed Cohen's d (59.1 for GA vs. B1, 16.3 for GA vs. B2) for continuity with the effect-size reporting convention, but flag that these values are not meaningfully interpretable using ordinary Cohen's d benchmarks at this sample size: with only five paired differences, d is dominated by the very small variance *among* those five differences rather than by the underlying effect in the way d is normally intended to summarise. Given the tightly clustered, uniformly positive raw differences, we treat the direction and approximate magnitude of the improvement over both baselines as well-supported by this analysis, while acknowledging that a larger number of seeds (e.g., 20–30) would still be preferable for a more conventional significance test and a tighter estimate of run-to-run variability; we list this alongside the other statistical recommendations in Section 5.8.

Table 6. Validation fitness across five independent runs (different random seeds), all evaluated under the same frozen scorer and stratified 80/20 validation split (seed 7) with final genomes archived per seed (Appendix B). GA = proposed optimised system; B1 = minimal-structure, short-budget baseline; B2 = fixed g_{bal}, all evaluated on the same fixed 400-prompt evaluation subsample (seed 7). One-sided Wilcoxon signed-rank tests give $W = 15$, $p = 0.031$ against both baselines; GA-vs-B1 differences span $[0.898, 0.933]$ (median 0.912; 95% CI $[0.904, 0.928]$; $d = 59.1$) and GA-vs-B2 differences span $[0.138, 0.161]$ (median 0.147; 95% CI $[0.141, 0.154]$; $d = 16.3$).

Run (seed)	GA Fitness	B1 Fitness	B2 Fitness
1 (13)	0.6993	−0.2065	0.5381
2 (26)	0.7027	−0.2090	0.5555
3 (39)	0.6999	−0.1984	0.5597
4 (52)	0.6996	−0.2318	0.5615
5 (65)	0.6949	−0.2380	0.5465
Mean ± Std	0.6993 ± 0.0028	-0.2168 ± 0.0172	0.5522 ± 0.0098

The GA fitness varies little across runs (std = 0.0028), but this figure is at the single-evaluation noise floor (SD ≈ 0.0027; Section 3.5), so it shows that the five converged genomes are indistinguishable to within measurement precision rather than providing a separate estimate of convergence stability. B1's much larger spread (std = 0.0172) is roughly six times that floor and does reflect genuine variability, consistent with its fitness being dominated by the response-part-budget truncation defect discussed in Section 5.5, whose effect on any given prompt sample is more sensitive to sampling than the well-optimised GA genome's is.These results show that the reported GA-optimised configurations consistently achieve higher fitness than the fixed seed reference B2 and the five individually sampled random genomes B3 under the same fixed generation and scoring path. This does not establish a causal advantage of evolutionary search alone, and the separate equal-budget random-search comparison in section 5.5shows no resolvable difference between methods under the present evaluation design.

4.8. Structure Component Usage

Early optimisation revealed structural collapse, in which responses relied heavily on questioning while omitting grounding and actionable guidance. After revising the structure scorer and response generator, all three structural components were consistently detected under the final scoring pipeline. Figure 6 shows the measured usage of grounding, action, and question components in the reported optimised configuration.

All three structural components reach a usage frequency of 1.0, indicating that the optimised genome ($w_c^\star = 0.768$) reliably inserts grounding, action, and question content into every response, consistent with the deterministic top regime ($w_c \geq 0.55$) described in Section 3.4.2.

All conditions reported in this manuscript were evaluated under one frozen scorer, with every run's final genome archived (Appendix B); the ordinal ranking B4 > B2 > B1 and the overall pattern of results are stable across all five significance-test seeds. The corrected marker lists were derived by inspecting responses on the same validation set used for reporting rather than a separate development set, so that split functions as a combined development and validation set rather than an untouched final test set (Section 5.8).

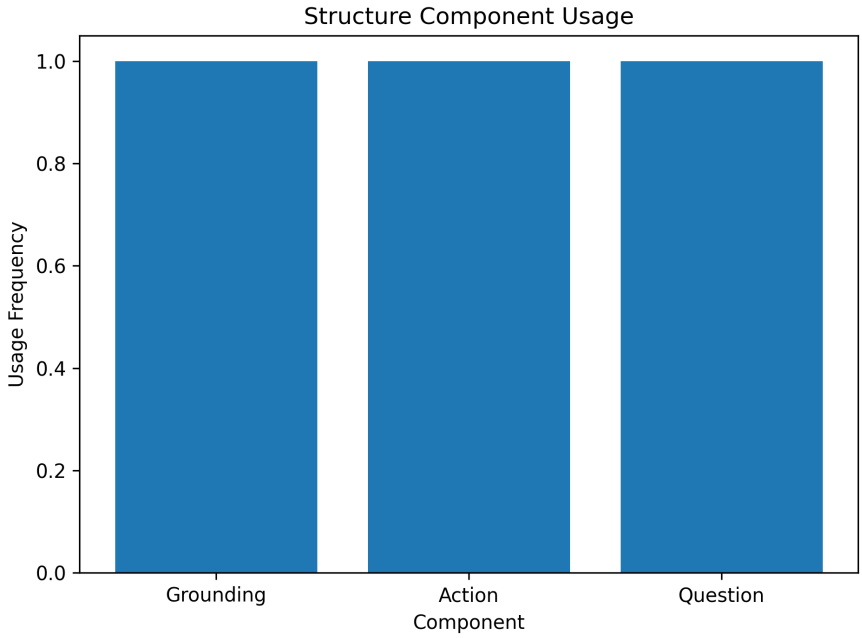


Figure 6. Structure component usage after revising the structure scoring function. Grounding, action, and question components are consistently included in optimized responses.

4.9. Risk-Aware Safety Behavior

Figure 7 shows safety scores across low-, mid-, and high-risk cases. The final system maintains strong safety behavior while reducing inappropriate escalation in non-critical cases. For high- and low-risk prompts, this risk-differentiation comes entirely from the fixed categorical safety-scoring rule applied per risk label (Section 3.5): escalation for high-risk prompts and its absence for low-risk prompts do not depend on any genome-evolved parameter. Within the mid-risk category specifically, escalation behaviour is not entirely fixed: the evolved genome parameter γ can in principle vary escalation with prompt content via the distress score h_i in the narrow band described in Section 3.4.3. We have not directly quantified, however, how many mid-risk prompts in this dataset actually fall in that band at $\gamma^\star \approx 0.059$. Figure 7's mid-risk safety distribution predominantly reflects the categorical lookup (Eq. (10)) applied to whichever prompts do or do not clear the fixed threshold $\tau_h = 0.65$, and we do not currently have direct evidence of a strong continuous trend driven by γ itself within the mid-risk category on this dataset.

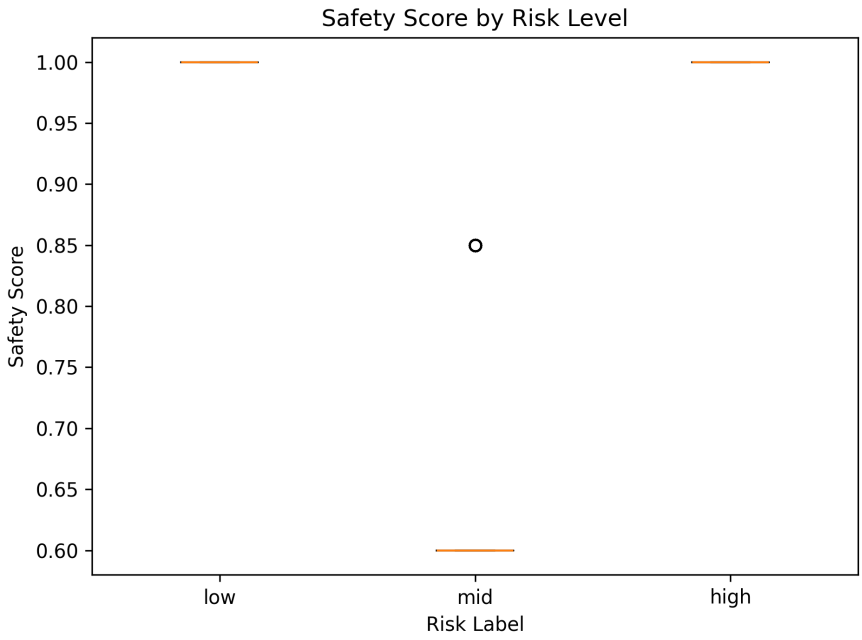


Figure 7. Safety score distribution by risk level.

4.10. Empathy–Structure Tradeoff

Figure 8 shows that the optimized system achieves high structural completeness while maintaining moderate empathy.

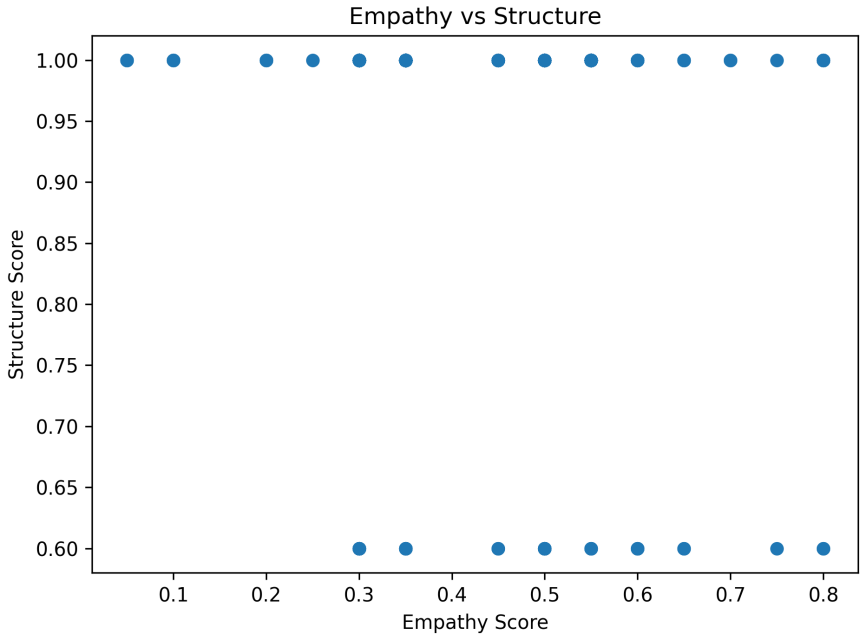


Figure 8. Empathy versus structure relationship.

4.11. Baseline vs. Optimised Response Comparison

To complement the quantitative baseline analysis in Section 4.7, Table 7 presents side-by-side qualitative comparisons of B1 (minimal-structure, short-budget baseline) and B4 (GA-optimised) responses across the three risk levels. Both columns are generated by running the actual `zeroshot_genome` and the reported GA genome ($w_c^\star = 0.768$, $w_e^\star = 0.131$, $w_s^\star = 0.101$) through the production response generator (Section 3.4) on matched prompts; the B1 examples shown are representative of the modal (most frequent) B1 output, since

B1's structural components are near-zero but not exactly zero across the validation set (Section 5.5). As documented in Section 5.5, the B1 responses shown here also illustrate a response-part-budget truncation defect, not only an unoptimised parameter setting; we recommend treating B1 primarily as a failure-mode illustration and B2 as the more informative non-GA comparator for judging the GA's own contribution (Section 5.5).

Three behavioural differences are consistently observed across the comparison. First, B1 responses are, in the modal case, structurally absent: no grounding instructions, no actionable guidance, and no reflective questions, consisting of an empathy line and a reflection line only, consistent with $w_c = 0.10$ placing B1 in the bottom, no-structure regime (Section 3.4.2). B1's structure score is correspondingly near-zero but not exactly zero (0.017, Table 5); we attribute the small residual to occasional edge cases in response assembly or scoring rather than to any deliberate structural inclusion. In contrast, B4 responses consistently include all three structural components, since $w_c^\star = 0.768$ falls in the top, deterministic regime, with each component serving a distinct therapeutic function: grounding exercises anchor the user in the present moment, action suggestions provide concrete next steps, and reflective questions invite further disclosure.

Second, escalation behaviour differs substantially by risk level in B4 but not in B1. The generator's escalation rule (Section 3.4.3) applies unconditionally to both whenever the risk label is high, independent of genome weight; the difference visible in Table 7 instead arises from B1's minimum response-part budget ($m = 256$) truncating escalation content before it reaches the visible response, whereas B4's $m = 512$ disables truncation (Eq. (7)). B4 accordingly suppresses explicit crisis language for low-risk prompts while reliably surfacing escalation guidance for high-risk ones. This risk-differentiation does not depend on θ_{mid} or θ_{high}, which the generator does not read; and the reported B1-to-B4 gains therefore combine GA optimisation with the effect of not triggering this defect, as decomposed in Section 5.5.

Third, B4 responses demonstrate multilingual composition, weaving Bangla, Banglish, and English phrases within a single response across functionally distinct components (grounding, action, and question); as with the prompts themselves (Section 3), this multilingual variation reflects the generator's hand-authored phrase banks rather than a claim about authentic code-switching patterns in real conversations. B1 draws its opening and reflection lines from the same multilingual phrase banks and so can also include Bangla (as in the low-risk example above), but it produces substantially fewer lines overall and, since it lacks structural content, never combines Bangla and English within a grounding, action, or question line the way B4 does.

These qualitative differences are consistent with the quantitative gaps reported in Table 5: structure differs by +0.983 (0.017 to 1.000), safety by +0.433 (0.472 to 0.905), and empathy by +0.161 (0.269 to 0.430). These differences correspond to observable differences in response content. However, the safety and structure differences in the B1–B4 comparison partly reflect B1's response-part-budget truncation defect rather than genetic optimisation alone.

The empathy gain is worth flagging on its own terms: the +0.161 absolute difference from B1 to B4 is smaller than the corresponding differences in structure (+0.983) and safety (+0.433). Thus, empathy is the smallest of the three automatic-metric differences in the B1–B4 comparison. The structure and safety differences should nevertheless be interpreted cautiously, because B1 also carries the short response-part-budget truncation defect discussed in Section 5.5. Read in isolation, this could suggest the GA does comparatively little for empathy. Section 4.12 complicates that reading: the same B1-vs-B4 comparison, rated blind by clinical psychologists, shows empathy improving by roughly as much as

Table 7. Side-by-side comparison of B1 (minimal-structure, short-budget baseline) and B4 (GA-optimised) responses across risk levels, both generated directly by the production pipeline.

Risk	**User Prompt (excerpt)**	**B1: Minimal-structure, short-budget**	**B4: GA-Optimised**
Low	"মনে হচ্ছে আমি একা হয়ে গেছি। I feel frustrated and angry a lot…"	"Feeling this alone can be really painful. তুমি একা না। It sounds like this has been weighing on you for a while."	"Feeling this alone can be really painful. তুমি একা না। চলো ধীরে ধীরে দেখি। চলো ৩০ সেকেন্ড grounding করি—গভীর শ্বাস নাও, তারপর ৫টা জিনিস দেখো। Try not to stay completely isolated with these thoughts. What would help you feel a little steadier in this moment?"
Mid	"I feel really anxious and overwhelmed. Family niye onek stress e achi…"	"I hear how much pressure you're carrying right now. It sounds like this has been weighing on you for a while."	"I hear how much pressure you're carrying right now. It sounds like this has been weighing on you for a while. এটা একা carry করতে হবে না। First, try to sit somewhere comfortable and take one slow breath. If possible, stay near someone who helps you feel calm. What would help you feel a little steadier in this moment?"
High	"I feel scared of what I might do if this gets worse…Can you help me stay safe right now?"	"I'm really sorry you're feeling this much pain right now. Your safety matters most. You might be carrying a lot emotionally right now."	"I'm really sorry you're feeling this much pain right now. Your safety matters most. You might be carrying a lot emotionally right now. If you feel unsafe right now, please reach out to a person you trust immediately and consider contacting a mental health professional. First, try to sit somewhere comfortable and take one slow breath. একজন trusted friend/family member-কে মেসেজ করা helpful হতে পারে। Is there someone nearby you feel comfortable talking to?"

safety and helpfulness. We return to this apparent conflict, and what it implies about the automatic empathy proxy, in Section 4.12.

4.12. Human Evaluation

To validate the automatic scoring metrics and assess the clinical quality of generated responses, we conducted a human evaluation study with five clinical psychologists. We note at the outset two design limitations that qualify the strength of this evaluation, discussed further in Section 5.8: first, the comparison evaluated is B1 (which, as documented in Section 5.5, carries a truncation defect rather than only unoptimised weights) versus B4, rather than the arguably more informative B2-versus-B4 comparison between a competently hand-configured system and the GA-optimised one; and second, the five participating psychologists also contributed to the risk-labeling design (Section 3.2.1) earlier in this project, so they are not fully independent of the system under evaluation in the way an external panel would be.

4.12.1. Evaluation Design

We sampled 30 response pairs from the baseline vs. optimised comparison dataset, stratified by risk level (6 low-risk, 9 mid-risk, 15 high-risk), stratified by risk level (6 low-risk, 9 mid-risk, and 15 high-risk items), approximately matching the corpus risk distribution. For each item, annotators were shown the user prompt and two responses labeled A and B in randomised order, without knowing which system generated each response (rater-blinded; the researchers preparing and analysing the response pairs were not themselves blinded to system identity, so we describe this design as single-blind from the annotators' perspective rather than double-blind). Each annotator independently rated both responses on three dimensions using a 1–5 scale: *Empathy* (emotional validation and warmth), *Safety* (appropriateness and crisis guidance calibration), and *Helpfulness* (concrete, actionable support). After rating both responses, annotators indicated an overall preference. Inter-annotator agreement was measured using Krippendorff's alpha [24] and pairwise percentage agreement within one scale point.

4.12.2. Results

Table 8 summarises the results. The GA-optimised system (B4) substantially outperforms the minimal-structure, short-budget baseline (B1) on all three dimensions, with item-level mean improvements of +1.68 on Empathy, +1.77 on Safety, and +1.95 on Helpfulness. These estimates and their significance tests are computed at the level of the 30 response-pair items rather than the 150 individual annotator-item ratings, since each item was rated by all five annotators and the within-item ratings are correlated rather than independent; we detail the analysis in Section 4.12.3 below. Across the 150 annotator-item preference judgements (5 annotators × 30 items), B4 was preferred in 146 (97.3%), B1 in 2 (1.3%), with 2 judgements (1.3%) marked equal. At the item level, all five annotators agreed unanimously on 27 of the 30 items (all favouring B4); the remaining three items each drew at least one dissenting judgement (one annotator preferring B1 on each of two items, and two annotators rating a third item as equal), rather than uniform agreement across all 30 items. Given that B1's disadvantage here partly reflects the truncation defect rather than parameter choice alone (Section 5.5), and that the comparison is not fully blinded to the study team, we read this result as showing that clinicians strongly prefer a complete, non-truncated, GA-optimised response over a truncated, minimally-weighted one, rather than as isolated evidence that evolutionary search specifically improves clinical quality beyond what a competently hand-configured system (B2) would already achieve; a B2-vs-B4 clinician comparison, which we did not run, would speak more directly to that latter question, and we list it as a required follow-up in Section 5.8.

These findings help interpret the tension flagged in Section 4.11 on the automatic proxy. In the B1–B4 comparison, the automatic empathy difference is 0.161, smaller than the corresponding structure and safety differences. On blinded clinician ratings of the same comparison, however, empathy improved by 1.68 points, similar in magnitude to the safety (1.77) and helpfulness (1.95) differences. We have not, however, formally tested whether these three effects are statistically equal to one another (e.g. via a system-by-dimension interaction test), so we describe them as similar in magnitude rather than as confirmed to be statistically indistinguishable.

The two measurements agree in direction, both favouring B4, but not in the apparent magnitude of the empathy improvement. One plausible explanation is limited sensitivity of the marker-based proxy: the rule-based empathy score saturates once a handful of distinct markers are present (Eq. (9)) and penalises repetition, so it has limited dynamic range to register further differences once a response clears that low bar. Clinicians reading the full response may still perceive and reward differences in warmth, specificity, and emotional attunement that the marker count does not distinguish. This is a concrete instance of the proxy-validity limitation discussed in Section 5.8: in this B1-vs-B4 comparison, the proxy may under-credit an improvement perceived by clinician raters, whereas the marker-stuffing case in Section 5.3 illustrates the opposite risk of over-crediting surface-level phrasing.

Table 8. Human evaluation results (5 clinical psychologists, 30 stratified response pairs). B4 = GA-optimised system; B1 = minimal-structure, short-budget baseline (see Section 5.5 for the truncation defect that qualifies this comparison). Scores on a 1–5 scale, shown as the mean and SD of all 150 individual annotator-item ratings (5 raters × 30 items) per condition. p-values are from a paired t-test at the item level ($n = 30$ independent items, each averaged across the 5 raters; see Section 4.12.3), not the raw $n = 150$ annotator-item ratings. α = Krippendorff's alpha (interval distance metric, computed with the `krippendorff` Python package; Section 4.12); pairwise agreement reports percentage of annotator pairs within 1 scale point.

Dimension	B4 Mean±SD	B1 Mean±SD	Diff.	p-value	α
Empathy	3.60 ± 0.82	1.92 ± 0.71	+1.68	2.8×10^{-16}	0.56
Safety	3.81 ± 0.77	2.04 ± 0.64	+1.77	1.4×10^{-22}	0.58
Helpfulness	3.69 ± 0.90	1.74 ± 0.53	+1.95	1.5×10^{-24}	0.61
Overall preference: B4 preferred in 97.3% of annotator-item judgements (146/150); B1 preferred in 1.3% (2/150); Equal in 1.3% (2/150).					

Descriptive safety scores are broadly similar across the three risk-stratified subsamples (B4 mean Safety: 3.80 low risk [$n = 6$], 3.76 mid risk [$n = 9$], 3.84 high risk [$n = 15$]); given the small per-stratum item counts, we describe this as showing no large descriptive differences across the three strata in this sample, rather than as confirming clinically appropriate escalation behaviour across all risk categories in general.

4.12.3. Correcting for Pseudoreplication

The 150 annotator-item ratings per condition are not 150 independent observations, since ratings are correlated within annotator and within item. We therefore report inference at the level of the 30 items, the genuine unit of independent replication, by averaging the 5 raters' scores per item and condition before computing paired differences. This yields the same mean improvements as the raw annotator-item means in Table 8 with correctly calibrated uncertainty, and remains highly significant under paired t-tests, Wilcoxon signed-

rank tests, and a crossed linear mixed model. Full statistics for all three checks are in Appendix F.

4.12.4. Inter-Annotator Agreement

We computed Krippendorff's alpha using the `krippendorff` Python package with an interval difference function over the 60 blinded response units (30 items × 2 responses), yielding a 5×60 reliability matrix with no missing ratings. Alpha was 0.56 for Empathy, 0.58 for Safety, and 0.61 for Helpfulness; percentile-bootstrap 95% confidence intervals from 2,000 resamples were $[0.48, 0.62]$, $[0.50, 0.64]$, and $[0.54, 0.66]$, respectively.

Krippendorff suggested $\alpha \geq 0.800$ when coded data are to be considered reliably interpretable by other scholars, and $\alpha \geq 0.667$ as a minimum for drawing only tentative conclusions; he also cautioned that these values are not universal cutoffs [25]. All point estimates and upper confidence bounds fall below 0.667. Accordingly, we interpret the 1–5 ratings as directional comparative evidence, rather than as calibrated absolute quality measurements.

Pairwise agreement within one scale point was 86.0% for Empathy, 87.2% for Safety, and 85.2% for Helpfulness, indicating that large rater divergences were uncommon. The preference results reported above provide separate evidence of a strongly directional B4-over-B1 preference; thus, we interpret the human evaluation as directional rather than as calibrated absolute quality measurement.

4.12.5. Validation of Automatic Metrics

Comparing Tables 5 and 8, the automatic empathy and safety scores rank B4 above B1 by the same ordering as the corresponding human ratings, on both dimensions. The two scales are not directly comparable in magnitude, but the consistent rank ordering supports the automatic scores as useful optimisation signals even without absolute calibration against clinical standards, keeping in mind the caveats above regarding B1's truncation defect and the partial lack of independence between the human evaluation panel and the system's design process.

5. Discussion

5.1. Role of Seeded Initialization

Seeded initialization plausibly plays a dual role in the proposed framework. On the positive side, seeding with four hand-designed genomes alongside randomly initialised individuals is a common approach for accelerating convergence and improving reproducibility across runs relative to fully random initialisation. Per-generation genome logs show that the near-optimal generation-1 best fitness (0.701, training-time) is attributable to a structure-heavy individual ($w_c \approx 0.68$) consistent with a mutated or crossed-over descendant of the structure-heavy seed g_{str}, rather than the balanced seed g_{bal}: g_{bal}'s own directly measured fitness (0.556 as B2) is well below that generation-1 value. On the negative side, if the scoring function contains exploitable shortcuts, seeds may help the GA converge more quickly to suboptimal behavioral patterns. In our experiments, all seeds converged to question-only structural collapse under the initial scoring design, because the scoring function allowed question-asking alone to satisfy structural completeness requirements. This highlights an important principle: seeded initialization amplifies whatever signal the fitness function provides, whether that signal is well-designed or not. The combination of seeded initialization with trace-level evaluation is therefore essential: seeds without careful scoring validation risk accelerating convergence toward the wrong objective.

The convergence pattern shows the GA reaching near-optimal training fitness in generation 1, via this structure-heavy lineage, and maintaining this through elitism while mu-

tation continues to explore, with no sustained second improvement phase, indicating the search does not discover a substantially better configuration than this early high-fitness region within 20 generations.

5.2. Importance of Trace Analysis

Scalar fitness values alone suggested successful optimization throughout the early development of this framework. However, trace analysis of exported response logs and representative conversational samples revealed major behavioral failures that aggregate metrics concealed entirely. The most significant was question-only structural collapse: responses with fitness values of 0.6–0.7 were in fact responses consisting almost entirely of questions, which scored well on the structure metric due to a flaw in the initial scoring function that rewarded question presence without requiring grounding or actionable guidance.

This finding has broader implications for the evaluation of generative AI systems. Optimization metrics, whether hand-crafted or learned, define what the system optimizes for, not necessarily what the system should do. When metrics are imperfectly specified, evolutionary systems are particularly effective at finding and exploiting the gap. Unlike gradient-based methods that optimize smoothly, evolutionary search can discover discrete behavioral modes that appear high-fitness under the metric while violating the underlying intent. Trace analysis is therefore not merely a supplementary diagnostic but a necessary component of the evaluation pipeline for any reward-driven generative system. The response-part-budget truncation defect documented in Section 5.5, and the marker-coverage gap documented in Section 4.8, are further instances of this same broader pattern: both were found only through tracing individual generated responses against internal generation flags, not through inspecting aggregate scores.

5.3. Scoring Function as Bottleneck

Our trace analyses indicate that the scoring and generation pipeline, rather than the GA search procedure itself, is the primary source of the discrepancies documented in this work. The first four failure modes illustrate reward hacking, in which an optimiser exploits an unintended shortcut in the stated objective rather than achieving the underlying intended behaviour [26]. In our context, we observed and either corrected or documented five distinct failure modes.

First, *question-only structural collapse*: the initial structure score rewarded question presence without requiring grounding or actionable guidance, leading the GA to generate responses consisting almost entirely of reflective questions. We corrected this by replacing the continuous, unbounded reward with the discrete component-count lookup in Eq. (11), under which a question-only response satisfies exactly one component and is capped at 0.20.

Second, *empathy marker stuffing*: the initial empathy score rewarded marker count linearly, incentivising responses that repeated validation phrases. We corrected this by introducing a stuffing penalty when total marker hits exceed 3.

Third, *safety saturation*: the initial safety score saturated quickly, providing little selection signal to distinguish adequate from stronger crisis responses. We introduced smooth safety soft thresholds, although the current threshold of 0.94 is unattainable for any mid-risk response (Section 3.5) and therefore does not fully resolve this limitation for that risk category.

Fourth, *structure-marker coverage gap*: an incomplete keyword list undercounted genuinely inserted structural components for a subset of high-risk responses (Section 4.8). We corrected this by expanding the marker lists.

Fifth, *response-part-budget truncation*: B1's minimal response-part budget silently drops escalation content appended late in the assembly order (Section 5.5). This is a generator implementation defect rather than a scoring-function issue per se, but we include it here because, like the preceding issues, it was detected through tracing individual responses rather than through aggregate metrics.

The interpretability of the genome and the modular scoring design enabled these diagnoses: each score component and generation step corresponded to a specific behavioural dimension, allowing failure modes to be attributed and, where possible, corrected individually. The first four issues concern the scoring or objective-design pipeline used the frozen scoring pipeline (Section 3.5.7). The fifth is specific to B1's construction rather than to the GA's scoring or search process, and is tracked separately as a required baseline correction (Section 5.8).

5.4. Data Efficiency and Practical Deployment

The dataset-size experiments, using nested training subsets and a pre-specified TOST equivalence test (Section 4.1), show that mean validation fitness stays within a narrow 0.0068-point band across all tested sizes ($N = 100$ to $N = 2297$) with no monotonic trend. TOST establishes statistical equivalence to the full-dataset condition at the pre-specified margin ($\varepsilon = 0.01$) for $N = 100$, $N = 300$, and $N = 1200$; only $N = 600$ falls marginally outside the margin, coinciding with the experiment's largest seed-level dispersion and not forming a monotonic pattern with subset size (Section 4.1). Mean runtime across the same nested-subset run ranges from 15.9 s at $N = 100$ to 478.2 s at $N = 1200$, without a monotonic trend (Figure 9); the guidance below therefore rests on fitness stability rather than on runtime scaling. Larger datasets provide no apparent performance benefit within the range tested.

For resource-constrained settings such as community mental-health NGOs or university counselling services in Bangladesh, these findings motivate testing whether useful calibration may be possible with datasets on the order of 100–300 de-identified support interactions, a scale achievable within a short deployment pilot. We frame this as a hypothesis for future testing rather than an established property. Equivalence at $\varepsilon = 0.01$ was demonstrated for $N = 100$ and $N = 300$ under the frozen scorer, but the margin is roughly three times the evaluation noise standard deviation (Section 3.5), the prompts are synthetic reconstructions, and the validation split also served scorer development; the result therefore concerns fitness stability under this proxy scorer rather than demonstrated calibration quality on real interactions. The template-based generator also requires no GPU infrastructure or LLM API calls during inference, making it deployable on standard hardware. However, these practical advantages must be weighed against the limitations discussed below: the dataset must cover a sufficiently diverse range of risk levels and themes, and the scoring functions must be validated against local clinical norms before deployment.

Fitness evaluation dominates runtime, so cost per GA run is approximately $\mathcal{O}(P \cdot T \cdot |\mathcal{I}|)$ — about 1.2M generate-and-score operations at $P = 100$, $T = 20$, $|\mathcal{I}| = 600$. Because $|\mathcal{I}|$ is capped at 600, this count does not grow with dataset size beyond $N = 600$. Measured wall-clock times do not show the corresponding plateau (Figure 9), which we attribute to timing variability on non-dedicated consumer hardware rather than to any algorithmic property; the total runtime (minutes, not GPU-hours) is the reliable takeaway, not its shape across N. This contrasts with fine-tuning or RLHF-based approaches, which typically require GPU infrastructure unavailable in the deployment settings this framework targets.

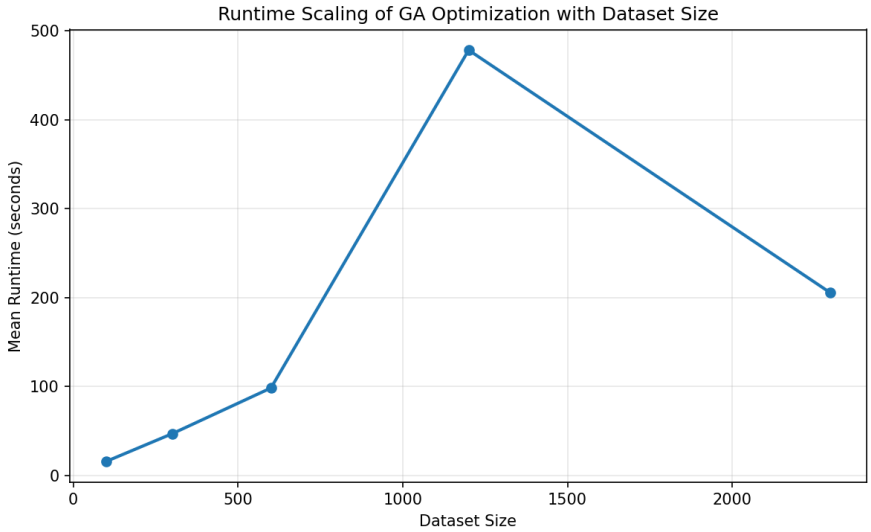


Figure 9. Runtime scaling of GA optimisation with dataset size (nested subsets, 5 seeds per size). Runtime does not scale monotonically with dataset size, peaking sharply at $N = 1200$ before falling at the full dataset size; see Section 5.4 for values and discussion.

5.5. Interpretation of Baseline Results

B1's weights ($w_e = 0.10$, $w_c = 0.10$, $w_s = 0.80$; Section 3) are chosen so that genome renormalisation leaves them unchanged, keeping w_c genuinely in the bottom, no-structure regime rather than drifting into the stochastic mid-regime a naive equal-floor construction would otherwise produce.

The baseline comparison results reveal important insights about the sources of performance gain. The near-zero structure score of B1 (0.013 in the common 400-prompt re-evaluation) shows that this minimal-weight configuration produces almost entirely unstructured responses, despite the underlying phrase banks containing grounding, action, and question components. Its negative fitness (-0.194) is primarily driven by low safety (0.475), which we trace to the response-part-budget truncation mechanism described in Section 3.4.1 (B1's $m = 256$ setting silently drops escalation content for high-risk prompts) rather than to failure of the fixed risk-aware escalation rule itself.

This is a genuine generator-design defect rather than merely an unoptimised parameter setting, because it can silently remove required crisis content. To isolate its effect, we re-evaluated B1, B1-fixed, and B4 using one frozen generation and scoring path on the identical fixed 400-prompt validation subsample. B1-fixed differs from B1 only in `memory_window` (512 rather than 256).

Table 9. Controlled truncation diagnostic. B1, B1-fixed, and B4 were re-evaluated using the same fixed generation and scoring path on the identical fixed 400-prompt validation subsample. B1-fixed differs from B1 only in `memory_window` (512 rather than 256). Values are from one common re-evaluation run and are interpreted as a diagnostic pattern rather than as noise-free effect estimates.

Condition	Empathy	Safety	Structure	Fitness
B1	0.273	0.475	0.013	−0.194
B1-fixed	0.323	0.905	0.012	0.277
B4	0.436	0.903	1.000	0.700

Table 9 shows that removing truncation raises Safety from 0.475 to 0.905 and Fitness from -0.194 to 0.277, while leaving Structure essentially unchanged at approximately 0.012. Empathy also rises from 0.273 to 0.323, suggesting that truncation removes response content

beyond escalation guidance. B4 achieves essentially the same Safety score (0.903), but increases Structure to 1.000, Empathy to 0.436, and Fitness to 0.700.

Thus, under this common evaluation, B1's safety deficit is consistent with being driven primarily by the response-part-budget truncation defect, whereas B4's structural advantage is not: it arises in the comparison between the non-truncating minimal-structure configuration and B4. The total fitness difference is descriptively divided between removal of truncation (+0.471 from B1 to B1-fixed) and the remaining B1-fixed–B4 configuration difference (+0.423). Because these values come from one common re-evaluation run, they should be interpreted as a controlled diagnostic pattern rather than precise noise-free effect estimates. The B1-fixed–B4 difference is not a pure estimate of the GA's causal contribution because the conditions differ in multiple genome parameters.

The B2–B4 comparison is more informative for assessing the remaining configuration difference, because B2 does not share B1's truncation defect. The balanced seed genome already achieves high safety (0.903) but leaves substantial room for structural improvement (0.648 vs. 1.000). The reported B4 configuration uses a high structure weight ($w_c^\star \approx 0.77$), which is associated with higher structural completeness than the balanced initialisation. We verified this directly with a 12-point grid search over w_c alone (holding the w_e:w_s ratio fixed at the reported genome's value, so that raw weights already sum to 1 and genome renormalisation is a no-op at every grid point): the results reproduce all three structure regimes exactly at the thresholds stated in Section 3.4.2 – structure is effectively zero for $w_c < 0.25$ (with small residual values possible because of response-assembly or scoring edge cases), fluctuates stochastically around 0.62–0.66 for $0.25 \leq w_c < 0.55$, and reaches 1.000 for every grid point at $w_c \geq 0.55$. Fitness within the deterministic regime is not flat: it rises from 0.652 at $w_c = 0.55$ to a peak of 0.710 at $w_c = 0.70$, then eases to 0.701 at the top of the range. The reported genome's $w_c^\star = 0.768$ therefore sits on the plateau but past its peak. This supports the interpretation that the structure-dominant result does not require evolutionary search specifically to discover: a twelve-point sweep over a single gene reaches the same fitness region as the GA's five converged runs, consistent with the generator's deterministic structure threshold ($w_c \geq 0.55$; Section 3.4.2) doing most of the explanatory work.

We additionally ran five independent equal-budget random searches (2,000 draws each over the full genome space – $w_e, w_s, w_c, \theta_{\text{mid}}, \theta_{\text{high}}, \gamma, d, m$ – matching the GA's implicit budget of $100 \times 20 = 2{,}000$ evaluations), sampled independently of the grid search above. Scored on the same 400-prompt subsample used for selection, the five best genomes reach 0.7066–0.7110 (mean 0.7085, SD 0.0018), above every GA seed in Table 6. That comparison is not sound: each random-search maximum is the largest of 2,000 noisy evaluations on the very set it is reported against, whereas the GA's genomes were selected on optimisation-set samples and never on this subsample. Taking a maximum over 2,000 draws from a measurement with SD ≈ 0.003 (Section 3.5) inflates the reported value by several standard deviations independently of genome quality.

Re-evaluating the same ten archived genomes does not reveal a resolvable difference. On the 575-prompt validation split, GA 0.7001 ± 0.0025 versus RS 0.7023 ± 0.0056; on the 400-prompt subsample, GA 0.7040 ± 0.0032 versus RS 0.7020 ± 0.0024. Neither set is fully clean for this comparison: the random searches selected on the 400-prompt subsample, which is contained in the 575-prompt split, so both are re-evaluations rather than held-out tests for RS. A re-score on the full 2,872-prompt corpus (GA 0.7015 ± 0.0019 versus RS 0.6996 ± 0.0013) is post-selection for RS but overlaps the GA's optimisation set, so it is not neutral in the other direction. Across all three sets the difference is ≈ 0.002 with an unstable sign, well inside the single-evaluation noise floor.

We therefore conclude that no difference between the two methods is resolvable under the present evaluation design, not that the two are equivalent; a genuinely untouched test set, evaluated only after selection for both methods, would be required to settle the question. Every random-search genome loses 0.007–0.012 relative to its archived selection-time score while the GA genomes do not, which is consistent with maximisation over noisy selection-time evaluations rather than demonstrating superior genome quality. This pattern is consistent with prior work on GA-based prompt optimisation, which reports selection bias when candidates selected on an evaluation set perform less well on a separate test set. In that study, a larger-population configuration also showed a larger test–validation gap, which the authors interpreted as a higher risk of overfitting; the authors further repeated reported evaluations because stochastic LLM outputs produced run-to-run score discrepancies [13].

The 2,000 draws underlying each random search have a mean fitness of only ≈ 0.33 (SD ≈ 0.33), so most random genomes are far from the plateau; what the five runs establish is that undirected search locates the plateau consistently at this budget, not that any individual draw is likely to.

The finding also suggests that safety and empathy are more easily satisfied by the fixed escalation rule, safety penalty terms, and phrase bank design than by genome weight allocation, freeing the evolutionary search to concentrate on structural completeness.

These observations answer RQ2 descriptively: B4 maintains safety comparable to B2 while substantially improving structural completeness. The grid search identifies the generator's deterministic structure threshold as the mechanism most plausibly responsible for the structural gain. We do not, however, attribute that gain causally to progressive structure enforcement in isolation: B2 and B4 differ in several respects simultaneously (genome weights, response-part budget, and γ), and ablations that vary each mechanism independently while holding the others fixed would be required to separate their contributions, as we note in Section 5.8.

5.6. Did the GA Achieve Multi-Objective Balance?

The fixed fitness-weight coefficients (Eq. (13)) assign empathy and safety the largest shares ($A = B = 0.40$) and structure the smallest ($C = 0.15$), reflecting our intended prioritisation of safe, empathetic responses. Yet the reported genome is structure-dominant ($w_c^\star = 0.768$ versus $w_e^\star = 0.131$; Table 4), and $w_s^\star$ is not part of this tension at all: as clarified in the genome note (Section 3), w_s is a renormalisation participant rather than an independently interpretable safety preference, since safety behaviour is governed by the fixed escalation rule and categorical safety lookup, not by w_s.

The resolution lies in distinguishing the *fixed* coefficients A, B, C, D, which set how base scores combine into fitness, from the *evolved* genome weights w_e, w_c, which determine how much empathy and structural content the generator actually inserts – these operate on different score landscapes. Safety and empathy both saturate quickly: safety reaches a high plateau ($S_i \approx 0.90$–1.00) once escalation is reliably present, independent of w_s, and empathy saturates once a few distinct markers appear, with further weight risking the marker-stuffing penalty. Structure, by contrast, has the largest unrealised headroom in the unoptimised baseline (B1 structure $= 0.017$) and responds sharply to w_c at the deterministic threshold (Section 3.4.2). This headroom asymmetry is compounded by an explicit reinforcement asymmetry in Eq. (13): structure is reinforced twice (up to $+0.09$ combined) while empathy is reinforced once (capped at $+0.05$) and safety not at all beyond its base term. So the structure-dominant outcome reflects both where headroom is largest *and* where the fitness function rewards marginal content most – neither of which is visible from the coefficient table A, B, C, D alone.

This suggests the GA found a corner solution – empathy near its saturation floor, structure maximised – rather than a smooth trade-off across objectives (Section 5.5: empathy stays modest while structure climbs substantially). This is a genuine limitation of collapsing multi-objective optimisation into a single scalar fitness with fixed coefficients: the resulting genome reflects the geometry of the constraint set as much as the analyst's stated priorities. A multi-objective evolutionary algorithm maintaining an explicit Pareto front over (E_i, S_i, C_i), rather than a weighted scalar sum, would make this trade-off space visible and let a clinician select an operating point directly; we list this as future work (Section 5.8).

5.7. Ethical Risks and Deployment Considerations

The deployment of optimisation-driven conversational agents in mental-health contexts raises ethical concerns that extend beyond algorithmic performance. This subsection critically examines deployment risks specific to the proposed framework and the Bangladeshi context.

Proxy metric misalignment. The empathy and safety scores are keyword-based proxies with the construct-validity limits detailed in Section 5.8; optimising against them risks producing responses that are *metrically safe* but not *clinically safe*. For deployment specifically, this means the proxies are not a substitute for clinical oversight: any deployment should include ongoing monitoring by qualified mental health professionals who can identify cases where high-scoring responses are clinically inappropriate.

Over-reliance and substitution risk. Systems like Riri are designed as supplementary tools, not replacements for professional care. However, the substitution risk is heightened by the scarcity of the alternative: Bangladesh has an estimated 1.17 mental health professionals per 100,000 population and no mental health services at primary-care or community level. The 2019 National Mental Health Survey found that more than 90% of adults and children with a mental health condition had not sought treatment, while participants in a qualitative stakeholder study reported delays of 6–13 years before seeking care, associated with stigma and low mental health literacy [27]. Users experiencing acute distress may therefore treat an AI assistant as a primary, or potentially their only, source of immediate support.

The high structural-component usage observed in our reported system (100% insertion rate) means that many responses include a referral to professional help where appropriate, but if no such help is accessible, this referral may feel dismissive rather than supportive. We also note that "100% structural-component usage" refers specifically to the generator's grounding/action/question components, not exclusively to professional referrals: an action component may equally be a breathing exercise, a suggestion to contact a trusted friend, or a grounding technique, and low-risk prompts specifically do not receive crisis escalation content at all (Section 3.4.3). Future work should design context-aware escalation that acknowledges service availability limitations.

Cultural and linguistic validity. The empathy marker sets used in this work were constructed by the research team from common Bangla/Banglish expressions observed in the dataset. They have not been validated by clinical psychologists or community mental-health workers familiar with Bangladeshi norms of emotional expression. Different communities within Bangladesh (urban versus rural and across socioeconomic backgrounds) may use different linguistic patterns to express distress; consequently, a marker set suited to the PHQ-4 screening population may not generalise. This is a documented failure mode rather than a hypothetical one: dominant sentiment-analysis models may misinterpret emotional restraint common in collectivist cultures as disengagement or low emotional involvement [6]. The screening instrument underlying our risk labels is on firmer ground:

the Bangla PHQ-4 has been psychometrically validated in rural and urban Bangladeshi community samples [17]. That validation was conducted in adult community samples and evaluated a screening cutoff, rather than the four-band severity stratification applied here to a self-referring digital-support population. This matters particularly because the reported genome's high structure weight ($w_c^\star = 0.768$) means structural components dominate the response, potentially at the cost of culturally appropriate emotional tone.

Reward hacking in deployment. The failure modes documented in this work (question-only collapse, empathy marker stuffing, the structure-marker coverage gap, and the response-part-budget truncation defect) were discovered through offline trace analysis before deployment. In a live deployment, new failure modes may emerge from interactions with real users whose language patterns differ from the training data. If the system is periodically re-optimised using interaction logs, the GA may discover new shortcuts that exploit patterns in real user behaviour. Robust deployment would require continuous trace monitoring, periodic human review of response samples, and clear mechanisms for clinical or designated safety reviewers to flag and correct problematic response patterns. The interpretable genome design facilitates this: because generator behaviour is controlled by a small set of named parameters, targeted corrections can be made without rerunning the full optimisation pipeline.

These considerations do not invalidate the proposed framework but rather define the conditions under which it could be responsibly deployed. The framework provides a technically informative, though not yet fully validated, exploration of low-resource mental-health conversational optimisation. Responsible deployment would require a broader governance structure including clinical oversight, ongoing human review of safety-relevant interactions, and culturally informed evaluation protocols [4–6].

Data provenance and risk-category validity. The high-risk category in our dataset is populated in part by PHQ-4 scores of 9–12 alone, independent of the explicit self-harm keyword indicator (Eq. (5)); because PHQ-4 has no suicide-risk item (Section 3.2.1), this may inflate the high-risk category with severe depression/anxiety presentations that do not necessarily involve suicidality, and reward crisis escalation for such cases without direct evidence of suicide risk specifically (Section 5.8).

Age field reliability and the 18+ platform policy. The application restricts registration to users aged 18 and over, and does not knowingly collect or use data from users under 18 for research. Age itself, however, is self-reported at registration and not independently verified (e.g. via identity document), and the dataset's age field (Appendix C, Table A2) shows 422 records (14.7%) with a self-reported age of 13–17. We cannot determine from the data alone whether these reflect genuine underage users who circumvented the registration policy, self-report errors, or another data-quality issue with this field, and we do not treat this figure as evidence that confirmed minors are present in the dataset. At the same time, unverified self-report cannot fully rule out their presence either, and we do not consider the platform's stated policy alone sufficient grounds to set the question aside. We recommend that future versions of this dataset either implement stronger age verification at registration, or apply an explicit data-governance rule that filters or separately reviews records with a self-reported age below 18 before they are used for research. We treat resolving this ambiguity as a prerequisite for any further research use of this dataset, including any onward sharing or reuse beyond the analyses reported here, and we report the present results together with the sensitivity analysis below so that readers can assess the extent to which they depend on the records in question.

To characterise how much this ambiguity affects the reported results specifically, we re-evaluated the already-trained genomes for B1, B1-fixed, B2, and B4 on the 575-prompt validation set with the 88 self-reported-minor records excluded (487 prompts), without re-

training anything, and compared against the same genomes' fitness on the full validation set (Table 10). This is a provisional analysis pending the institutional determination described above, and should not be read as resolving the underlying governance question – it addresses only whether the currently reported numbers are statistically sensitive to this subset, not whether these records should be included in the study at all.

Table 10. Sensitivity of reported fitness to excluding the 88 self-reported minor-age records from the 575-prompt validation set. "Diff/SD" is the absolute difference between full and minor-excluded evaluations, computed from unrounded fitness values and divided by the condition's known run-to-run evaluation SD. B1-fixed has no repeated-evaluation SD available. Differences exceeding approximately 1.5× the known SD are flagged as exceeding normal evaluation noise.

Condition	Full validation (n=575)	Minors excluded (n=487)	Absolute difference / run-to-run SD	Assessment
B1	−0.216	−0.223	0.40×	within normal noise
B1-fixed	0.284	0.287	n/a	no comparison SD available
B2	0.549	0.571	2.30×	exceeds normal noise
B4	0.698	0.696	0.26×	within normal noise

The headline GA result (B4) is essentially unaffected: its fitness shift (0.698 to 0.696) is well within the genome's own normal run-to-run variability. B2's shift (0.549 to 0.571), however, exceeds its known evaluation noise by more than double, indicating a genuine, non-trivial sensitivity to this subset for that specific baseline; we do not have a confirmed mechanism for why B2 is more sensitive to this subset than B4 or B1 are, and we do not speculate further than noting the observation. We read this as evidence that the reported headline result is robust to this question, while explicitly flagging that not every reported condition is equally robust, and we do not treat this sensitivity check as a substitute for resolving the underlying governance question.

5.8. Limitations and Future Work

The following limitations define the scope of the present findings. Detailed diagnostic evidence is reported in the referenced sections; here we summarise the implications for interpretation and the corresponding priorities for future work.

Data, labels, and governance. The prompts evaluated here are synthetic reconstructions from structured theme fields rather than verbatim user messages. Consequently, the findings describe behaviour on this synthetic corpus and do not establish ecological validity for authentic Bangla, Banglish, or code-switched mental-health conversations. The low-, mid-, and high-risk categories combine PHQ-4 severity with a keyword override and are not validated suicide-risk classifications. In addition, age is self-reported and a subset of records is labelled 13–17 years; resolving whether these records reflect genuine underage users or age-field error, and applying an explicit governance decision, is required before further research use of the dataset (Section 5.7).

Evaluation and causal attribution. The validation split also contributed to scorer development, so it should be understood as a combined development–validation set rather than an untouched final test set. B1 contains a response-part-budget truncation defect, making it primarily a failure-mode reference rather than a clean unoptimised baseline. Although B2 provides the most informative fixed non-GA reference, B2 and B4 differ in multiple configuration settings; their comparison therefore does not isolate the causal contribution of

evolutionary search. The B1-fixed truncation diagnostic is based on a single re-evaluation and should be read as a diagnostic pattern rather than as a precise causal-effect estimate. Likewise, no evaluation set is untouched by both the GA and equal-budget random-search selection procedures, so the present study cannot resolve whether either method has a general advantage. B5 was evaluated in one run only and requires repeated MSGA-initialised runs for a reliable comparison. Finally, the nested dataset-size and population-size experiments establish equivalence at the pre-specified TOST margin for all tested population sizes and for three of the four tested dataset sizes, but on five seeds per condition and with a margin of $\varepsilon = 0.01$ that is roughly three times the single-evaluation noise standard deviation; the result bounds the size of any effect rather than showing that the conditions are identical.

Measurement and human evaluation. Generation is stochastic, giving repeated evaluations of the same genome an observed fitness standard deviation of approximately 0.003. Differences near this scale, including the B4–B5 and GA–random-search differences, are not resolvable under the present evaluation design. The dataset-size and population-size differences are of this same order, and each condition contributes only five independent GA seeds; the equivalence tests reported for them therefore bound the size of any difference at the pre-specified $\varepsilon = 0.01$ margin, which is roughly three times that noise standard deviation, rather than resolving the individual differences or showing the conditions to be identical. The baseline significance tests use only five GA seeds: their one-sided Wilcoxon result establishes directional consistency across those runs, but does not provide a precise estimate of population-level uncertainty. Reported population "diversity" is fitness dispersion rather than direct genome-space diversity. The automatic empathy and safety scores are keyword-based proxies that cannot fully capture contextual appropriateness, emotional attunement, sarcasm, negation, or culturally specific expression. The clinician evaluation compares B1 with B4 rather than B2 with B4; its raters also contributed to the earlier risk-label design, and inter-rater agreement supports directional preference rather than calibrated absolute quality claims. Future work should use an independent panel, an anchored rating rubric, and a B2–B4 comparison.

Scope and deployment. The framework evaluates single-turn, template-based responses rather than multi-turn dialogue. It therefore cannot assess longer-horizon risks such as emotional dependence, reassurance loops, belief reinforcement, or changes in risk across a conversation. The current mid-risk safety rule applies a penalty even when its highest attainable safety score is reached, and some genome genes are inactive or only weakly identified by the present objective. The reported policy is also not personalised: one fixed genome is applied to all users, without modelling individual language preferences or preferences for directive versus reflective support. Future work should evaluate authentic conversational data under appropriate governance, use independent test sets and repeated seeded generations, validate the metrics with local clinical and community stakeholders, and extend the objective to multi-turn, clinician-audited safety outcomes.

Natural extensions of this work include:

1. Multi-turn sequential genome policies, in which the GA optimises escalation and structural decisions conditioned on conversation history. This direction is motivated by work arguing that mental-health chatbot safety should be evaluated at the dialogue level and that risk-relevant effects can accumulate across extended interactions [18]. It may also provide selection pressure on γ, which the present single-turn objective cannot identify (Appendix D.1).
2. A more robust generator design that separates response-length truncation from internal template and safety-control tokens, or reorders response assembly so that escalation content is inserted before lower-priority content and cannot be discarded by

the response-part budget. This would prevent the class of defect illustrated by B1 by construction, rather than merely diagnosing it through comparison with B1-fixed.

3. Multi-objective evolutionary algorithms that maintain explicit Pareto fronts over empathy, safety, and structure, making trade-offs visible to a clinician rather than collapsing them into a single scalar fitness.
4. Learned empathy and safety evaluators trained and validated on independently clinician-annotated data, replacing or complementing the current rule-based proxy scores.
5. Human-feedback-informed evaluation, in which independently collected clinician and user ratings are used to validate or refine the fitness function, with safety-critical criteria retained separately rather than reduced to a single preference score.
6. An independent clinician-in-the-loop evaluation, including B2 and a non-truncating B1 variant alongside B4, with mental-health professionals rating blinded response pairs under a controlled evaluation protocol. Any subsequent live deployment study would require separate clinical oversight, governance, and safety monitoring.
7. Multilingual expansion beyond Bangla/Banglish/English to other South Asian languages where mental-health support infrastructure is similarly limited.

6. Conclusions

We presented a risk-aware genetic optimization framework for a mental-health conversational agent operating in Bangla, Banglish, and English. By evolving an interpretable genome controlling a structured generator, the system achieves substantial improvements in response structure and overall fitness, maintaining an already-strong escalation-calibration safety proxy and modestly increasing empathy relative to a fixed, hand-designed baseline genome. All results come from a single, consistent scoring pipeline with per-seed genomes archived throughout. Decomposing the largest baseline gap (B1 to B4) shows it splits between a software-level response-truncation effect and genuine genome optimisation in different proportions for safety and structure respectively (Section 5.5); a grid search and five equal-budget random searches over the genome space reach the same fitness region as the GA, indicating the structure-dominant result reflects a broadly reachable fitness plateau for this genome space and fitness design rather than something evolutionary search was specifically required to find (Section 5.5). Final fitness is statistically equivalent to the full-dataset condition for the tested nested subsets $N = 100$, $N = 300$, and $N = 1200$, but not for $N = 600$, at the pre-specified equivalence margin of ± 0.01 fitness points. Final fitness is statistically equivalent across all tested population sizes from $P = 30$ to $P = 400$ relative to $P = 100$ (Sections 4.1 and 4.2). The study illustrates how evolutionary algorithms can optimize non-differentiable, value-laden objectives, and highlights the role of scoring design, trace-based evaluation, and careful baseline construction in reporting AI systems for mental-health support. The framework's observed low-data performance stability, its interpretability, and its deployment feasibility on standard hardware make it a promising foundation for resource-constrained mental-health applications in multilingual settings, within the scope and limitations detailed in Section 5.8.

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Institutional Review Board Statement: The dataset used in this study consists of de-identified prompts collected as part of routine mental-health screening and support services in

Bangladesh. The application restricts registration to users aged 18 and over, and does not knowingly collect or use data from users under 18 for research. Users provided informed consent within the application at the point of data collection, acknowledging that anonymised interaction data may be used for research purposes. No personally identifiable information was retained, and the prompts analysed in this study are synthetic reconstructions from structured summary fields rather than verbatim user text (Section 3). Research use of the de-identified data was authorised by the data controller's internal research team, which determined that secondary analysis of fully anonymised service-delivery data did not require external ethics review. We note explicitly that this determination was made internally by the organisation that operates the application, and that the corresponding author is affiliated with that organisation (see Conflicts of Interest); it was not issued by an independent institutional review board or ethics committee. The study was conducted in accordance with the Declaration of Helsinki. As noted in Section 5.7, age at registration is self-reported and not independently verified, and a subset of records (14.7%) carry a self-reported age of 13–17; we cannot determine from the data alone whether this reflects genuine underage users or self-report error. We treat resolving this ambiguity, via stronger age verification or explicit filtering of under-18 self-reports, as a prerequisite for any further research use of this dataset, and we report a sensitivity analysis (Section 5.7) showing that the headline result reported here is not materially affected by excluding the records concerned.

Informed Consent Statement: Informed consent was obtained from all participants via the mental-health support application at the point of data collection. Users were informed that their interaction data would be anonymised, and acknowledged that anonymised interaction data may be used for research purposes. No personally identifiable information was retained and no individual participants can be identified from the published data.

Data Availability Statement: The dataset used in this study is not publicly available due to patient privacy restrictions and the sensitive nature of mental-health screening data. The source code for the EvoRiri framework, including the genetic algorithm, response generator, and scoring functions, is available from the corresponding author upon reasonable request.

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Conflicts of Interest: The corresponding author's PhD supervisor, Prof. Mir Md. Jahangir Kabir, serves as Guest Editor of the Special Issue "Evolutionary Algorithms in Artificial Intelligence" in *Mathematics* to which this manuscript is submitted. This conflict of interest has been disclosed to the MDPI editorial office prior to submission in accordance with MDPI's conflict of interest policy. Additionally, the corresponding author is affiliated with Relaxy PTE. LTD., which owns and commercially develops *Riri*, the system evaluated in this manuscript. *Riri* is provided to users at no cost, and this research received no dedicated funding or commercial sponsorship; it was conducted with the goal of identifying cost-effective methods to improve *Riri* for its user base. This affiliation and product relationship are disclosed here in accordance with MDPI's conflict of interest policy.

Appendix A. Evolutionary Failure Mode Examples

This appendix provides concrete examples of the two evolutionary failure modes documented in Section 5: question-only structural collapse and empathy marker stuffing. These failure modes were discovered through trace analysis during early development, before the scoring function was redesigned. The examples below are representative of the response patterns produced by high-fitness genomes under the initial (flawed) scoring design.

Appendix A.1. Question-Only Structural Collapse

Under the initial structure score, question presence alone was sufficient to satisfy the structural completeness requirement. Genomes with high question weight converged to

responses that scored 1.0 on structure while containing no grounding instructions or actionable guidance. A representative example:

> *User:* "I feel really anxious and overwhelmed. Family niye onek stress e achi. What should I do now?"
>
> *Failed response (pre-fix, high structure score):* "How long have you been feeling this way? What do you think is causing the most stress? Have you talked to anyone about this? Is there something specific that happened recently? What usually helps you feel better?"
>
> *Structure score:* 1.0 *Empathy score:* 0.05 *Safety score:* 0.60

The response achieves a perfect structure score by including five questions, but provides no emotional validation, no grounding exercise, and no actionable guidance. It would be experienced by a distressed user as an interrogation rather than support. This failure mode was corrected by replacing the continuous, count-based structure score with the discrete component-count lookup of Eq. (11), under which any response satisfying only one of the three components (grounding, action, or question) is capped at 0.20, regardless of how many times that single component is echoed.

Appendix A.2. Empathy Marker Stuffing

Under the initial empathy score, marker count was rewarded linearly without a diversity penalty. Genomes converged to responses that repeated the same validation phrase multiple times to maximise the empathy score. A representative example:

> *User:* "Ami onek lonely feel kortesi lately. I feel really anxious and overwhelmed."
>
> *Failed response (pre-fix, high empathy score):* "It makes sense that you feel this way. It makes sense to feel overwhelmed. It makes sense that this is hard. It makes sense that you're struggling. What would help you right now?"
>
> *Empathy score:* 0.90 *Structure score:* 0.20

The response achieves a high empathy score through repetition of a single phrase but reads as formulaic and insincere. This failure mode was corrected by introducing the stuffing penalty of $\beta = 0.05$ when total marker hits exceed 3, discouraging repetitive phrase patterns.

Appendix A.3. Post-Fix Optimised Response (Same Prompt)

After scoring redesign, the same prompt from Appendix A.2 produces:

> *Optimised response (post-fix):* "Feeling this alone can be really painful. তুমি একা না। First, try to sit somewhere comfortable and take one slow breath. Try not to stay completely isolated with these thoughts. Is there someone nearby you feel comfortable talking to?"
>
> *Empathy score:* 0.50 *Structure score:* 1.0 *Safety score:* 0.60 (mid-risk, no escalation content; see Appendix A.4 for the corresponding fitness calculation)

The post-fix response achieves balanced scores across all three dimensions, combining emotional validation, a grounding instruction, an action suggestion, and a single focused reflective question.

Appendix A.4. Worked Example: From Prompt to Fitness

This subsection walks the same prompt and response from Appendix A.3 through every stage of the pipeline in Figure 1, using the reported genome $g^\star$ from Table 4 ($w_e^\star = 0.131$, $w_s^\star = 0.101$, $w_c^\star = 0.768$, $\theta^\star_{\text{mid}} = 0.588$, $\theta^\star_{\text{high}} = 0.700$).

1. **Prompt + risk label.** x_i = "Ami onek lonely feel kortesi lately. I feel really anxious and overwhelmed." For illustration we take $r_i = \text{mid}$.
2. **Genome.** The generator is configured with $g^\star$ as above (Eq. (6)).
3. **Generator.** Because $w_c^\star = 0.768$ falls in the top structure regime ($w_c \geq 0.55$; Section 3.4.2), the generator deterministically includes grounding, action, and question components ($c_g = c_a = c_q = 1$); $w_e^\star$ governs how many empathy lines are included.
4. **Response.** $y_i = G(x_i, r_i; g^\star)$ = the response shown in Appendix A.3 above.
5. **Automatic scoring.** Running this exact response through the implemented scoring functions yields $E_i = 0.50$, $S_i = 0.60$, $C_i = 1.00$ (Eqs. (9)–(11)). The response contains no escalation-language marker (Section 3.5), which is expected and appropriate for a mid-risk prompt whose computed distress score is low ($h_i \approx 0$, well below $\tau_h = 0.65$); under the categorical safety lookup (Eq. (10)), a mid-risk response without escalation content scores $S_i = 0.60$ by construction, and $S_i = 1.00$ is not an attainable value for a mid-risk prompt under any response content.
6. **Fitness.** The response is below the 140-word penalty threshold, so $L_i = 0$. Since $S_i = 0.60 < 0.94$, the flat safety soft-threshold penalty of -0.15 applies (the risk label is mid, not high, so the -0.50 high-risk penalty does not apply); since $h_i \leq \tau_h$, the distress-gating penalty (Eq. (13), final line) also does not apply. $C_i = 1.00 \geq 0.70$, so the structure soft penalty does not apply; $E_i = 0.50 < 0.62$, so the empathy soft penalty does apply. The per-example fitness (Eq. (13)) is

$$\begin{aligned} f_i(g^\star) &= \underbrace{0.40(0.50) + 0.40(0.60) + 0.15(1.00)}_{\text{base } AE_i+BS_i+CC_i} - \underbrace{0.15}_{\text{safety soft-threshold penalty}} - \underbrace{0.18\,\tfrac{0.62-0.50}{0.62}}_{\text{empathy soft penalty}} \\ &\quad + \underbrace{0.05(0.50) + 0.05(1.00) + 0.05\min(1.00, 0.8)}_{\text{reinforcement terms}} \\ &\approx 0.590 - 0.150 - 0.035 + 0.025 + 0.050 + 0.040 = 0.520. \end{aligned}$$

   This illustrates the per-example fitness computation (Eq. (13)) that underlies both the training-time fitness $F_{\text{train},t}(g) = \frac{1}{600}\sum_{i\in\mathcal{I}_t} f_i(g)$, averaged over the $|\mathcal{I}_t| = 600$ prompts sampled from the optimisation set at generation t (Eq. (14)), and the validation fitness $F_{\text{val}}(g) = \frac{1}{400}\sum_{i\in V_{400}} f_i(g)$, averaged over the fixed 400-prompt subsample of the separate, validation set. The reported validation fitness $F_{\text{val}}(g^\star) \approx 0.703$ (Table 5) is computed on this validation subsample, not on any single generation's 600-prompt training sample; the two quantities use different, non-overlapping prompt populations and are not directly interchangeable, though both use the same per-example fitness formula. The single-example value above is lower than the reported $F_{\text{val}}(g^\star)$ because this example's response does not contain any escalation content and so incurs the safety soft-threshold penalty, whereas across the validation set most low- and high-risk prompts (which make up the majority of the dataset; Appendix C) attain $S_i = 1.00$ under Eq. (10) and so do not incur this penalty.
7. **GA feedback.** Across generations, genomes whose responses score well on this fitness are more likely to be selected, crossed over, and mutated into the next population (Section 3.6), which is how the search arrived at $g^\star$ from the seeded initial population (Appendix B).

Appendix B. Optimised Genome Evolution Examples

Appendix B.1. Renormalisation, w_s, Reserved Escalation Genes, and the Omitted Length-Penalty Gene

The three continuous weights w_e, w_s, w_c are stored in raw (post-mutation, pre-normalisation) form in the genome, and are renormalised to sum to 1 at the point the

generator reads them, after all crossover and mutation for that individual has been applied. Consequently, two genomes with identical raw w_e and w_c but different raw w_s will be assembled into *different* responses, because the shared normalising denominator $w_e + w_s + w_c$ differs between them. w_s's evolved value is therefore best interpreted as a normalisation-only participant rather than as an independently meaningful "safety weight," and we avoid describing $w_s^\star = 0.101$ as reflecting an evolved safety preference (Section 5.6); its only demonstrated effect on any reported result is through the denominator it contributes to.

θ_{mid} and θ_{high} are two further genes, part of the genome and subject to crossover and mutation like every other gene, but in the present implementation not read by the generator or the fitness function, and therefore with no effect on any reported result. We retain them in the genome because they are the intended interface for graded, distress-score-driven escalation, listed as future work in Section 5.8.

The genome as implemented does not include a separate evolvable length-penalty gene: the length penalty in Eq. (13) uses the fixed coefficient $D = 0.05$ throughout, rather than a per-genome value. An earlier design considered introducing such a gene (denoted λ), but it was not carried into the implementation, so we omit it from Eq. (6) rather than list a gene the genetic algorithm never actually evolves.

Appendix B.2. Genome Evolution Across Generations

The reported genome (Table 4) scores 0.703, just above the band spanned by the five independent significance-test seeds (Table 6: 0.6949–0.7027). We describe it as the "reported genome" rather than the "best genome" because we performed no formal top-1 selection across the six runs conducted, and because a 0.003-point margin is inside the single-evaluation noise floor (Section 3.5). Table A1 shows representative genome configurations at three stages of the GA run: the balanced seed initialisation, an intermediate generation, and the final converged genome.

Table A1. Representative genome configurations at three GA stages, from the main run conducted under the frozen scoring pipeline (Section 3.5.7). w_e, w_s, w_c are empathy, safety, and structure weights (renormalised to sum to 1). θ_{mid}, θ_{high} are reserved escalation-threshold genes that are evolved alongside the others but are not currently read by the generator or fitness function (Section 3.4.3); included here for completeness. Note that "Fitness" here is training-time fitness on a per-generation $|\mathcal{I}| = 600$ sample, not the validation-set fitness reported for B2 in Table 5; the two are not directly comparable (Section 4).

Stage	w_e	w_s	w_c	θ_{mid}	θ_{high}	**Fitness**
Seed g_{bal} (Gen 0)	0.330	0.330	0.340	0.550	0.800	0.532
Intermediate (Gen 10)	0.147	0.102	0.751	0.689	0.700	0.702
Reported genome (Gen 19)	0.131	0.101	0.768	0.588	0.700	0.704

The evolution shown here is for the main run and shows a directional shift within that run: w_c increases from 0.340 to 0.768, while w_e decreases toward its minimum floor value; w_s settles at a moderate raw value above its floor, though as discussed in the genome note (Section 3) this raw value's only demonstrated effect is via the shared normalising denominator, not as an independently meaningful safety preference. This reflects the fitness landscape discussed in Section 3.8: safety is largely satisfied by the fixed escalation rule and safety penalty terms, and empathy by the phrase bank design, leaving structural completeness as the primary differentiator between genomes on this run. Whether this same structure-dominant configuration is robustly discovered across independent runs, rather

than being distinctive to this one, has direct supporting evidence: the five independent seeds run for the significance analysis (Table 6) converge to a narrow band of final validation fitness (0.6949–0.7027), all well above g_{bal}'s 0.556 and close to the 0.703 reported here, indicating that high final fitness (and, by the same headroom argument as Section 5.6, high structure weight) is a robust outcome of this search rather than an artefact of one run.

By contrast, θ_{mid} and θ_{high} show no comparable directional trend across the same three stages (0.550→0.689→0.588 and 0.800→0.700→0.700 respectively): they move under mutation without the monotonic pressure visible in w_c. This is consistent with Section 3.4.3: because these two genes are not read by the generator or the fitness function in the present implementation, they carry no selection gradient and are expected to drift under mutation rather than converge, which is exactly the pattern observed here.

Appendix B.3. Identifying the Generation-1 Best Individual

Per-generation genome logs from the main GA run show the generation-1 best individual has $w_c \approx 0.678$, $w_e \approx 0.132$, $w_s \approx 0.190$ – clearly *not* g_{bal} (whose structure weight is $w_c \approx 0.34$), and instead already structurally similar to the eventual reported genome ($w_c^\star = 0.768$). This is consistent with the generation-1 population containing a mutated or crossed-over descendant of the structure-heavy seed g_{str} (or, less likely, a favourably-drawn random individual) rather than g_{bal}.

Appendix C. Dataset Composition

Table A2 summarises the composition of the full $N = 2872$ prompt dataset used in all experiments, recomputed from the underlying PHQ-4 screening data (`phq4_cleaned.csv`) after filtering out rows with no recorded theme content (407 of 3,279 collected rows).

Table A2. Dataset composition ($N = 2872$ de-identified prompts). PHQ-4 severity bands: Normal (0–2), Mild (3–5), Moderate (6–8), Severe (9–12). Risk labels are derived from PHQ-4 total score and explicit self-harm/suicide indicator detection (Section 3).

Category	Count	Proportion
Risk label distribution		
High risk	1,342	46.7%
Mid risk	833	29.0%
Low risk	697	24.3%
PHQ-4 severity distribution		
Severe (9–12)	1,283	44.7%
Moderate (6–8)	871	30.3%
Mild (3–5)	494	17.2%
Normal (0–2)	224	7.8%
Language of interaction		
Bengali (Bangla)	2,745	95.6%
English	120	4.2%
Banglish	7	0.2%
Age group		
18–22 years	1,054	36.7%
23–27 years	696	24.2%
13–17 years	422	14.7%
28–34 years	418	14.6%
35–44 years	187	6.5%
45+ years	95	3.3%
Gender		
Male	2,063	71.8%
Female	716	24.9%
Other	93	3.2%

Several aspects of the dataset composition are noteworthy for interpreting the experimental results. First, the dataset is heavily skewed toward high-risk cases (46.7%), reflecting the clinical context of the application; as discussed in Section 5.7, part of this high-risk category reflects PHQ-4 severity alone rather than confirmed suicide/self-harm risk specifically. Second, the dataset is predominantly young (60.9% aged 18–27), male (71.8%), and Bengali-speaking (95.6%), which limits generalisability of the empathy marker sets to other populations. Third, the high proportion of severe PHQ-4 scores (44.7% scoring 9–12) confirms that the dataset captures genuine clinical distress rather than mild wellbeing concerns, though, again, not necessarily suicide-specific risk.

Appendix D. Distress Scoring and Penalty Terms

Appendix D.1. The Direct Distress-Gating Penalty Term

Equation (13) includes a further term, $-0.10\,\mathbb{I}\{h_i > \tau_h \wedge r_i = \text{mid} \wedge \text{esc}_i = 0\}$, that penalises a mid-risk response lacking escalation content when h_i exceeds the *fixed* threshold

τ_h (not the γ-relaxed effective threshold). Because the effective threshold $\tau_h/(1+\gamma) \leq \tau_h$ for any $\gamma \geq 0$, whenever $h_i > \tau_h$ the escalation condition in Eq. (8) is already satisfied regardless of γ; a correctly functioning, untruncated generator should therefore always set $\mathsf{esc}_i = 1$ in this regime, and $\mathsf{esc}_i = 0$ here can only arise from a downstream generation failure (e.g. the response-part truncation described in Section 3.4.1 and analysed for B1 in Section 5.5) rather than from the evolved value of γ itself. This penalty term therefore does not supply direct selection pressure on γ. The region in which γ's value actually changes escalation behaviour is $\tau_h/(1+\gamma) < h_i \leq \tau_h$, where $h_i \leq \tau_h$ and the direct penalty above does not apply; any selection pressure on γ in that region operates only indirectly, through the resulting change in the categorical safety score between $\mathsf{esc}_i = 0$ (0.60) and $\mathsf{esc}_i = 1$ (0.85) for mid-risk prompts whose distress score falls in that narrow band (Eq. (10)). We have not separately quantified how many mid-risk prompts in the dataset fall in this band at $\gamma^\star \approx 0.059$, nor run a $\gamma = 0$ ablation holding all other genes fixed, so we do not assert a magnitude of benefit we have not directly measured.

Appendix D.2. The Marker-Stuffing Penalty on Distress Scores

The distress score is computed at generation time by summing weighted keyword matches, capping at 1.0, and applying a small penalty for marker stuffing (>3 distinct signals). This penalty is clinically questionable on its own terms: discounting a score for matching "too many" markers is appropriate for the empathy-marker-stuffing failure mode of Section 5.3, where markers reflect the *generator's* output and repetition signals gaming, but the same logic does not transfer to a *user's* distress signal, where several distinct indicators (hopelessness, isolation, and self-harm language together) plausibly warrant more concern rather than less.

We verified computationally that the penalty is nonetheless inert here. Its trigger condition (more than three distinct matched keywords) is never satisfied on any prompt entering the fitness pipeline — the concatenated `common_themes` / `thought_progression` text that `compute_distress_score()` receives — where the maximum number of distinct matches is two. (Clinician `risk_assessment` notes can match four or more if scanned as free text, but those are used only for risk-label construction, not as the distress-score input.) No escalation decision reported in this paper is affected. This does not resolve the underlying design concern: a scorer that cannot discount more severe presentations would be preferable in future revisions.

Appendix E. Equivalence Tests and Sensitivity Statistics

Appendix E.1. Equivalence Testing Procedure

Equivalence was assessed with Schuirmann's two one-sided tests (TOST). The primary analysis treats the two conditions as *independent samples* and uses the Welch unequal-variance t statistic with Welch–Satterthwaite degrees of freedom, at a pre-specified margin of $\varepsilon = 0.01$ fitness points and $\alpha = 0.05$. Independent-samples Welch TOST is appropriate here because each condition is run with its own five GA seeds and the conditions differ markedly in seed-level dispersion (standard deviations from 0.0016 to 0.0073), so equal variances cannot be assumed.

Equivalence requires rejecting both one-sided nulls, $H_0^- : \Delta \leq -\varepsilon$ (an upper-tail test) and $H_0^+ : \Delta \geq +\varepsilon$ (a lower-tail test), where Δ is the difference in mean validation fitness between a condition and its reference. The reported TOST p-value is $\max(p_{\text{lower}}, p_{\text{upper}})$. Intervals are reported at the $(1-2\alpha) = 90\%$ level, which is the level corresponding to the TOST decision: equivalence holds exactly when the 90% confidence interval lies entirely inside $(-\varepsilon, +\varepsilon)$. A 95% interval is *not* the correct reference interval for a TOST decision at $\alpha = 0.05$ and is not reported for these tests. Each condition contributes five independent

GA seeds, scored under the frozen pipeline (Section 3.5.7). The implementation depends on SciPy only; all nine tests were independently reproduced with a second, established implementation as a numerical check.

Table A3. TOST equivalence tests for dataset size (against the full-dataset condition $N = 2297$) and population size (against $P = 100$). Independent-samples Welch TOST, margin $\varepsilon = 0.01$ fitness points, $\alpha = 0.05$, five GA seeds per condition. Intervals are 90% confidence intervals, the level matching the TOST decision at this α. The TOST p-value is $\max(p_{\text{lower}}, p_{\text{upper}})$, and the final column reports whether equivalence at $\varepsilon = 0.01$ is established (i.e. whether TOST $p < \alpha$). Mean differences are given to five decimal places and interval bounds to four; p-values are given to three significant figures.

Comp.	Mean diff.	90% CI	p_{lower}	p_{upper}	TOST p	Equiv.?
Dataset size, nested subsets (reference $N = 2297$)						
$N = 100$	+0.00122	$[-0.0051, +0.0075]$	0.00615	0.0168	0.0168	Yes
$N = 300$	−0.00198	$[-0.0067, +0.0027]$	0.00678	7.69×10^{-4}	0.00678	Yes
$N = 600$	+0.00475	$[-0.0024, +0.0119]$	0.00340	0.101	0.101	No
$N = 1200$	+0.00290	$[-0.0008, +0.0066]$	3.34×10^{-4}	0.00527	0.00527	Yes
Population size (reference $P = 100$)						
$P = 30$	+0.00111	$[-0.0024, +0.0046]$	5.02×10^{-4}	0.00146	0.00146	Yes
$P = 50$	−0.00000	$[-0.0045, +0.0045]$	0.00332	0.00331	0.00332	Yes
$P = 200$	+0.00210	$[-0.0002, +0.0044]$	7.17×10^{-6}	1.26×10^{-4}	1.26×10^{-4}	Yes
$P = 300$	−0.00117	$[-0.0051, +0.0028]$	0.00304	0.00107	0.00304	Yes
$P = 400$	+0.00253	$[-0.0012, +0.0063]$	4.39×10^{-4}	0.00461	0.00461	Yes

Appendix E.2. Dataset-Size Equivalence Testing

Three of the four nested subset sizes are statistically equivalent to the full-dataset condition at $\varepsilon = 0.01$ (Table A3). $N = 600$ is the exception, failing the upper one-sided test ($p_{\text{upper}} = 0.101$) while passing the lower test ($p_{\text{lower}} = 0.00340$): its mean is 0.0048 points above $N = 2297$ and its 90% confidence interval reaches +0.0119, marginally beyond $+\varepsilon$. That condition also carries the largest seed-level standard deviation in the experiment (0.0073, against 0.0017 at $N = 1200$), and the four tested sizes do not form a monotonic pattern: both the smallest and the largest subsets tested meet the margin. We report these observations without claiming to have established why $N = 600$ behaves differently, and we do not generalise the equivalence result to untested subset sizes: the claim applies to $N = 100$, $N = 300$, and $N = 1200$ as tested.

Appendix E.3. Population-Size Equivalence Testing

All five population sizes are statistically equivalent to $P = 100$ at $\varepsilon = 0.01$ (Table A3). Raw mean differences are smaller than in the dataset-size experiment (all at or below 0.0025 fitness points, versus a 0.0068-point band across dataset sizes), and the widest 90% confidence interval is $[-0.0051, +0.0028]$ at $P = 300$, so no interval approaches the margin. Mean runtime, by contrast, ranges from 64 seconds at $P = 30$ to a non-monotonic peak of 882 seconds at $P = 200$, so the computational cost of larger populations buys no fitness gain detectable at this margin.

Appendix F. Human Evaluation Statistical Details

Appendix F.1. Correcting for Pseudoreplication

The 150 annotator-item ratings per condition (5 annotators × 30 items) are not 150 independent observations: ratings from the same annotator are correlated across items, and ratings of the same item are correlated across annotators. Treating all 150 as independent,

as in a naive paired t-test on the raw annotator-item pairs, understates the true uncertainty and can overstate significance. We therefore report inference at the level of the 30 items, the genuine unit of independent replication in this design, rather than at the level of individual ratings.

For each dimension, we first average the 5 raters' scores for each item and condition, then compute the paired difference (B4 minus B1) across the resulting $n = 30$ independent item-level pairs. This yields a mean item-level improvement of +1.68 (Empathy), +1.77 (Safety), and +1.95 (Helpfulness), matching the raw annotator-item means in Table 8, but now with correctly calibrated uncertainty: 95% confidence intervals of $[1.48, 1.88]$ (Empathy), $[1.64, 1.89]$ (Safety), and $[1.83, 2.06]$ (Helpfulness), obtained both from the normal approximation and from a 10,000-resample percentile bootstrap over items, which agree closely with each other. Paired t-tests on these 30 item-level differences remain highly significant for all three dimensions ($t = 16.51$, $p = 2.8 \times 10^{-16}$ for Empathy; $t = 28.07$, $p = 1.4 \times 10^{-22}$ for Safety; $t = 32.96$, $p = 1.5 \times 10^{-24}$ for Helpfulness), as do Wilcoxon signed-rank tests run on the same item-level differences as a nonparametric check ($p < 10^{-5}$ in all three cases). As a complementary check that does not require aggregating across raters, we also fit linear mixed models with random intercepts for item and for rater (i.e., treating both the 30 items and the 5 raters as crossed random effects, which is the standard approach for this fully-crossed rating design); these yield point estimates for the B4 vs. B1 effect that are numerically identical to the item-level analysis (+1.68, +1.77, +1.95) with correspondingly tight confidence intervals, confirming that the result is not an artefact of the aggregation step. We report the item-level analysis as primary in Table 8 because it depends on fewer distributional assumptions and is directly reproducible by hand from the 30 item-level means. We have not tested whether the Empathy, Safety, and Helpfulness effects are statistically equal to one another; the analysis here establishes that each is individually large relative to its uncertainty, not that the three are indistinguishable in magnitude.

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